

BOARD OF INTERMEDIATE EDUCATION KARACHI · BIEK

Computer Science I & II

Lecture Notes

Class XI+XII · Science General & Humanities · Printed lectures for college and self-study

Sindh Curriculum 2019

BIEK Model Paper 2026

English + Urdu

C / VB / MS Access

What is in this book

Paper	Computer Science Paper I · 75 marks · 3 hours
Section A	15 MCQs · 15 marks · OMR
Section B	Short answers · 30 marks
Section C	Detailed answers · 30 marks · diagrams required
Units	Computer system, memory, system unit, OS, networks & data communication, plus Sindh programming bridge

Paper	Computer Science Paper II · 75 marks · choose ONE option
Option I	Programming using C + Database / MS Access
Option II	Programming using Visual Basic + Database / MS Access
Also covered	Pointers, OOP (C++), file handling, SDLC, multimedia, wireless — Sindh Grade XII units

Unofficial teaching notes for students of BIEK-affiliated colleges. Definitions, programs and diagrams follow the Sindh HSC Computer Science curriculum and the official BIEK paper pattern (Section A MCQs, Section B short, Section C detailed). Turbo C / void main() style is shown because that is what Karachi board answer books still expect.

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Unit C4: Programming using C

22. Selection: if, if-else, nested if and switch ★

Unit C5: Programming using C

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Unit 2: Pointers (Sindh XII + C paper)

26. Pointers and the address operator ★

Unit 3–4: OOP using C++ and file handling

27. OOP in C++: class, object, constructor, inheritance, polymorphism ★

Unit VB1: Programming using Visual Basic

28. Visual Basic IDE, forms and ActiveX controls ★

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29. VB language: operators, if, Select Case, loops, arrays, functions ★

Unit 5: Database fundamentals

30. DBMS, models, ER diagrams, keys and normalisation ★

31. MS Access objects, data types and SQL ★

Unit 1: System Development Life Cycle

32. SDLC, stakeholders and phases ★

Unit 6–7: Multimedia and wireless

33. Multimedia systems; wireless and mobile communication

Unit R: Revision

34. Paper II Option I mock (C + Access) with answers ★

35. Paper II Option II mock (VB + Access) with answers ★

BIEK Paper I pattern and how to study

Read once, then return before every test · How to use these lectures

What you will learn

- State the BIEK Computer Science I paper structure and marks
- Know which units carry weight in Sindh Curriculum 2019
- Write Section B answers in 6–8 lines with a definition + points + example
- Draw the diagrams that examiners expect in Section C

Who this book is for

You are a Class XI student of a Karachi college affiliated with the Board of Intermediate Education, Karachi (BIEK). Computer Science Paper I is the theory paper of HSC Part I (Science General or Humanities). These lectures follow the Sindh Curriculum for Computer Science Grades XI–XII (2019) and the official BIEK Model Paper 2026.

LEARN IT

Paper I is 75 marks in 3 hours. It is NOT the programming paper. Programming in C or Visual Basic is Paper II in Class XII. Paper I tests hardware, software, memory, CPU, operating system, data communication and networks.

Section	What you do	Marks	Time (typical)
A — MCQs	15 questions, all compulsory, on OMR	15	20 min
B — Short	Short notes, differences, full forms	30	1 h 20 min
C — Detailed	Long answers with diagrams	30	1 h 20 min

Sindh unit weightage (Grade XI)

Unit	Title	Weight
1	Overview of Computer System	15%
2	Computer Memory	10%
3	Inside System Unit	15%
4	Operating System	10%
5	Programming Concept Using C++	20%
6	Arrays, String and Structure	15%
7	Computer Communication & Networks	15%

HOW BIEK ASKS THIS

BIEK Model Paper 2026 Paper I asked: IT uses/abuses, barcode reader, hardcopy vs softcopy, virus, plotter, fetch cycle, modulation, compiler vs interpreter, impact vs non-impact printers, analog vs digital, OSI model, OS functions, registers, secondary storage, network classification. Memorise those exact headings.

REMEMBER IT

Section B method: (1) one-line definition, (2) 3–4 labelled points, (3) one example. Section C method: definition, labelled diagram, 5–7 explained points, comparison table if asked.

Write full forms when an abbreviation appears: CPU, RAM, ROM, OSI, SMTP, BIOS.

اردو میں سمجھیں

یہ کتاب بورڈ آف انٹرمیڈیٹ ایجوکیشن کراچی (BIEK) کے کمپیوٹر سائنس پیپر ون کے لیے لکچر نوٹس ہے۔ پیپر ون میں ہارڈویئر، سافٹ ویئر، میموری، آپریٹنگ سسٹم اور نیٹ ورک آتے ہیں۔ پروگرامنگ کلاس بارہویں کے پیپر ٹو میں ہے۔ سیکشن بی میں تعریف، تین نکات اور ایک مثال لکھیں۔ سیکشن سی میں لیبل والا خاکہ لازمی بنائیں۔

COMMON BOARD MISTAKE

Do not write Python answers in a BIEK Paper I / Paper II C paper. Karachi board still examines C (Option I) or Visual Basic (Option II) plus MS Access.

CHECK YOURSELF

1. How many MCQs are in Paper I Section A?

2. Which paper contains C programming?

Answers: 1. 15 (one mark each) · 2. Paper II (Class XII), Option I

MCQs

1. BIEK Computer Science Paper I total marks are:

- a. 50
- b. 60
- c. 75
- d. 100

2. Programming using C is offered in:

- a. Paper I only
- b. Paper II Option I
- c. Paper II Option II
- d. Practical only

Key: 1=c, 2=b

Short questions (3 marks each)

1. Write the scheme of BIEK Computer Science Paper I.
2. Why must Section C answers include a diagram?

Model short answers

1. 75 marks, 3 hours: Section A 15 MCQs (15), Section B short (30), Section C detailed (30). All of Paper I is theory of computer systems, OS and networks.
2. Detailed questions ask you to explain a model (OSI, fetch cycle, topologies). Marks are given for a labelled sketch plus explanation. A paragraph without a figure loses marks.

Computer, IT and the system block diagram ★ Golden

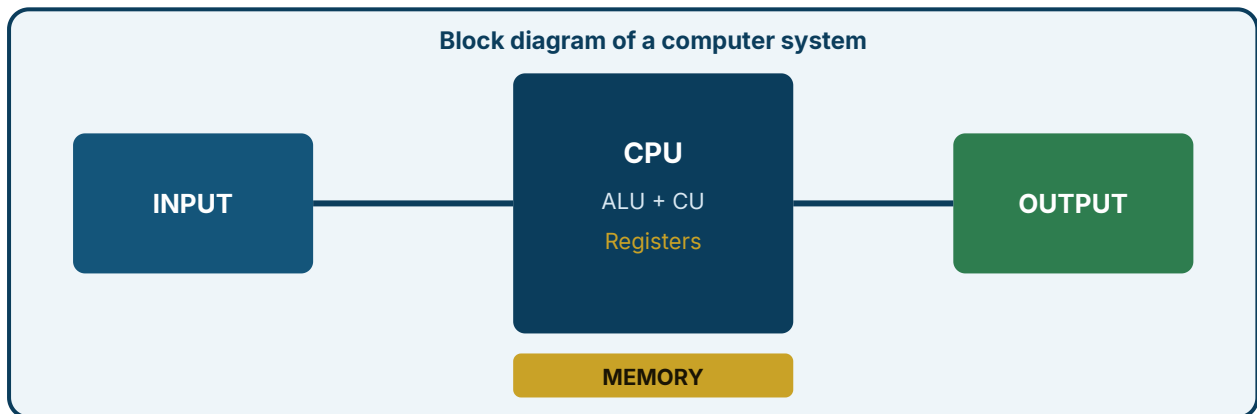
One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Define computer and Information Technology
- Draw the block diagram: input, CPU, memory, output
- List uses and abuses of IT
- State the four basic operations: input, process, output, storage

Computer. An electronic data-processing machine that accepts data (input), processes it according to instructions, produces information (output) and stores results.

Information Technology (IT). The use of computers, networking and software to collect, store, process, protect and transmit information.



IPO model: Input → Process (CPU + Memory) → Output. Storage sits under the CPU.

LEARN IT

Every computer does four operations. Input brings data in. Processing is done by the CPU (ALU calculates, CU controls). Output shows results. Storage keeps data in memory or on disk.

Hardware is the physical parts you can touch. Software is the set of programs that tell hardware what to do. Firmware (BIOS) is software stored in ROM.

Uses of IT

- Education: LMS, online classes, digital libraries.
- Business: billing, inventory, e-commerce, ATM / online banking.
- Health: MRI, patient records, telemedicine.
- Government: NADRA, tax, e-office, SECCAP admissions.
- Communication: email, video call, social media.

Abuses of IT

- Hacking and identity theft.
- Computer virus and ransomware.

- Cyber-bullying and fake news.
- Privacy loss and excessive screen time.
- Software piracy and plagiarism.

WORKED EXAMPLE / BOARD STYLE

Board short question: "What is Information Technology? Write any three uses or abuses of IT."

Answer: IT is the application of computers and telecommunication to handle information. Uses: online banking, hospital records, distance learning. Abuses: hacking, viruses, spread of false information.

HOW BIEK ASKS THIS

BIEK 2026 Paper I Q2(i) is exactly this heading. Keep a 5-line ready answer plus 3 uses and 3 abuses.

REMEMBER IT

IPO memory trick: In-Process-Out, and Store underneath. Draw arrows; examiners look for arrows.

اردو میں سمجھیں

کمپیوٹر ایک برقی مشین ہے جو ڈیٹا لیتی ہے، ہدایات کے مطابق اس پر عمل کرتی ہے، نتیجہ دیتی ہے اور محفوظ کرتی ہے۔ انفارمیشن ٹیکنالوجی کمپیوٹر اور نیٹ ورک سے معلومات کو جمع، محفوظ، پراسیس اور منتقل کرنے کا نام ہے۔ فائدے تعلیم، صحت، بینکنگ میں ہیں؛ نقصان وائرس، ہیکنگ اور جعلی خبریں ہیں۔

CHECK YOURSELF

1. Name the four basic operations of a computer.
2. Who performs processing?

Answers: 1. Input, process, output, storage · 2. CPU (ALU + CU + registers)

MCQs

1. The primary function of the CPU is:
 - a. To input data
 - b. To store data
 - c. To process data
 - d. To print data
2. The brain of the computer is the:
 - a. RAM
 - b. Hard disk
 - c. CPU
 - d. Keyboard
3. IT stands for:
 - a. Internet Tool
 - b. Information Technology
 - c. Internal Transfer
 - d. Input Terminal

Key: 1=c, 2=c, 3=b

Short questions (3 marks each)

1. Define computer. Draw its block diagram.
2. Write three uses and three abuses of IT.

Model short answers

1. A computer is an electronic device that inputs, processes, outputs and stores data. Diagram: Input devices → CPU (ALU, CU, registers) with Memory below → Output devices.
2. Uses: education, e-banking, hospitals. Abuses: hacking, viruses, fake news / privacy loss.

Long questions (board Section C)

1. What is a computer system? Explain hardware, software, basic operations and memory with a labelled block diagram.

Hardware: input, output, storage and communication devices

★ **Golden**

One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Classify devices as input, output, storage, processing or communication
- Describe keyboard, mouse, scanner, barcode, OCR, MICR, OMR, joystick, microphone, webcam
- Describe monitors, printers, plotter, speakers, projector
- Differentiate hardcopy and softcopy; impact and non-impact printers

Hardware. The physical, electronic and mechanical parts of a computer system.

Class	Job	Examples
Input	Send data into the computer	Keyboard, mouse, scanner, MICR, OMR, OCR, barcode, joystick, microphone, webcam, stylus, touch screen
Output	Send results to the user	Monitor, printer, plotter, speaker, projector, headphones
Storage	Keep data permanently or temporarily	HDD, SSD, USB, SD, CD/DVD/Blu-ray
Processing	Transform data	CPU, GPU
Communication	Send data between devices / networks	NIC, modem, switch, hub, router, gateway, fax

Input devices the board loves

- Barcode reader: optical scanner that reads product bars in a superstore for price and stock.
- OCR (Optical Character Recognition): reads printed text into editable text.
- OMR (Optical Mark Recognition): reads filled bubbles on MCQ sheets.
- MICR (Magnetic Ink Character Recognition): reads cheque numbers in banks.
- Scanner / data tablet / joystick / voice recognition / touch screen / webcam / microphone.

Output: hardcopy vs softcopy

	Hardcopy	Softcopy
Meaning	Printed permanent copy	Temporary display on screen
Device	Printer, plotter	Monitor, speaker, projector
Example	Marksheet on paper	Result on the college portal

Printers

Impact (hit the paper)	Non-impact (no hit)
Dot matrix, daisy wheel, line printer	Inkjet, laser, thermal
Noisy, carbon copies possible	Quiet, high quality
Cheap per page for multipart forms	Laser is fast for offices

WORKED EXAMPLE / BOARD STYLE

Plotter: an output device that draws large maps, CAD drawings and banners with pens. Board 2026 asked: "What devices are useful to print huge drawing or maps?" Answer: plotter (drum or flatbed).

HOW BIEK ASKS THIS

Ready differences: Hardcopy/Softcopy; Impact/Non-impact (four points); Barcode use in a superstore.

COMMON BOARD MISTAKE

A mouse is input, not output. A plotter is output, not a printer type you use for ordinary pages. Oracle is a DBMS, not an OS.

اردو میں سمجھیں

بارڈویئر چھونے والے حصے ہیں۔ ان پٹ کی بورڈ، ماؤس، بار کوڈ ریڈر، آؤٹ پٹ مانیٹر، پرنٹر، پلاٹر۔

بارڈ کاپی کاغذ پر چھپی کاپی ہے، سافٹ کاپی اسکرین پر۔ امپیکٹ پرنٹر کاغذ کو ٹکر مارتے ہیں، نان امپیکٹ (لیزر، انک جیٹ) نہیں مارتے۔

CHECK YOURSELF

1. Is OMR an input or output device?
2. Which printer makes carbon copies?

Answers: 1. Input — it reads marks from paper · 2. Impact / dot-matrix

MCQs

1. A scanning input device is:
 - a. Keyboard
 - b. Mouse
 - c. Track ball
 - d. OMR
2. Floppy disk and hard disk are types of:
 - a. Optical disk
 - b. Magnetic tape
 - c. Magnetic disk
 - d. Flash memory
3. A plotter is used to print:
 - a. Letters
 - b. Huge drawings and maps
 - c. Only photos
 - d. Sound

Key: 1=d, 2=c, 3=b

Short questions (3 marks each)

1. How can a barcode reader be useful in a superstore?
2. Differentiate hardcopy and softcopy.

3. Write any four differences between impact and non-impact printers.
4. What devices print huge drawings or maps?

Model short answers

1. It scans the bar code, fetches price and stock from the database, speeds billing and reduces typing errors.
2. Hardcopy is printed paper (permanent); softcopy is on screen or speaker (temporary, easy to edit).
3. Impact: strike paper, noisy, carbon copies, cheaper for forms. Non-impact: no strike, quiet, high quality, laser/inkjet.
4. Plotters (and large-format printers).

Long questions (board Section C)

1. Classify computer hardware. Explain input and output devices with examples.

Software, translators, virus and cutting-edge technology

★ Golden

One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Distinguish system, application and utility software
- Explain compiler, interpreter, assembler and compare compiler with interpreter
- Define computer virus and name antiviruses
- Outline AI and cloud computing at Grade XI level

Software. A set of programs, procedures and documentation that instruct the hardware.

Type	Role	Examples
System software	Runs the computer; manages resources	OS (Windows, Linux, macOS, Android), device drivers, BIOS
Language translator	Converts source code to machine code	Compiler, interpreter, assembler
Application software	User jobs	MS Word, Excel, Chrome, Turbo C IDE, Photoshop
Utility	Maintenance	Antivirus, compression (WinRAR), disk cleanup, backup, data recovery

Translators

	Compiler	Interpreter
How	Translates the whole program first, then runs the object code	Translates and runs one statement at a time
Speed	Faster at run time	Slower at run time
Errors	Shows a list of errors after compilation	Stops at the first error
Object file	Creates .exe / .obj	Usually no separate object file
Example	C, C++	Python, BASIC (classic)

Assembler translates assembly language (mnemonics) into machine code. Source code is the program you write; object code is the machine-language result.

Computer virus

LEARN IT

A computer virus is a malicious program that attaches itself to other files, replicates, and can damage data, steal information or slow the system. Worms spread on networks without a host file. Trojans pretend to be useful software. Ransomware locks files until a ransom is paid.

- Examples of viruses/malware: ILOVEYOU, WannaCry, Trojan horses, spyware, adware.
- Antiviruses: Kaspersky, Avast, AVG, Norton, Bitdefender, Windows Defender, McAfee, Avira.

Cutting-edge technology (Unit 1.5)

- Artificial Intelligence (AI): making machines perform tasks that need human intelligence — speech, vision, expert advice. Applications: face unlock, chatbots, medical imaging, self-driving research.
- Cloud computing: using servers on the internet (Google Drive, Microsoft 365) so you need less local storage. Advantages: access anywhere, backup, pay-as-you-go, easy sharing.

HOW BIEK ASKS THIS

2026 Paper I: define virus with examples and name five antiviruses; compiler vs interpreter; HTML as the markup language for web pages.

COMMON BOARD MISTAKE

Windows is system software; MS Word is application software. A virus is software, not hardware. HTML is a markup language, not a full programming language like C, but BIEK MCQs still key it as the language of web pages.

اردو میں سمجھیں

سافٹ ویئر پروگراموں کا مجموعہ ہے۔ سسٹم سافٹ ویئر کمپیوٹر چلاتا ہے، ایپلی کیشن صارف کا کام کرتی ہے، یوٹیلیٹی صفائی اور حفاظت کرتی ہے۔

Compiler پورا پروگرام ایک ساتھ مشین کوڈ بناتا ہے، Interpreter لائن بہ لائن۔ وائرس نقصان دہ پروگرام ہے جو نقل کرتا ہے؛ اینٹی وائرس اس سے بچاتا ہے۔

CHECK YOURSELF

1. Name three language translators.
2. Give two advantages of cloud computing.

Answers: 1. Compiler, interpreter, assembler · 2. Access from anywhere; automatic backup / less local hardware

MCQs

1. A computer virus is a:
 - a. Signal
 - b. Software
 - c. Hardware
 - d. Firmware
2. Windows is an example of:
 - a. Application software
 - b. System software
 - c. Utility only

d. Compiler

3. The markup language used to produce web pages is:

- a. XML
- b. HTML
- c. SGML
- d. WML

4. C programs are converted into machine language by a:

- a. Editor
- b. Assembler
- c. Compiler
- d. Debugger

Key: 1=b, 2=b, 3=b, 4=c

Short questions (3 marks each)

1. How is a compiler more efficient than an interpreter?
2. Define computer virus with examples. Name any five antiviruses.
3. Differentiate system software and application software.

Model short answers

1. A compiler translates the whole program once into object code which then runs fast. An interpreter translates every line each time the program runs, so it is slower and stops at the first error.
2. A virus is malicious replicating software (examples: worms, Trojans, ransomware). Antiviruses: Kaspersky, Avast, Norton, Bitdefender, Windows Defender.
3. System software controls hardware (OS, drivers). Application software does user tasks (Word, browser, Turbo C).

Long questions (board Section C)

1. Explain types of software. Discuss language translators with a comparison of compiler and interpreter.

Memory units, RAM, ROM and measurement ★ Golden

One college period · 40–45 minutes · Computer Memory

What you will learn

- State bit, nibble, byte and the binary prefixes KB–TB
- Differentiate volatile and non-volatile memory
- Explain RAM (SRAM, DRAM) and ROM (PROM, EPROM, EEPROM)
- Compare primary and secondary memory

Unit	Size
Bit	0 or 1 — smallest unit of data
Nibble	4 bits
Byte	8 bits — stores one character
KB (Kilobyte)	1024 bytes
MB (Megabyte)	1024 KB
GB (Gigabyte)	1024 MB
TB (Terabyte)	1024 GB

RAM

volatile · read/write
holds running programs
SRAM / DRAM

ROM

non-volatile · mostly read
holds BIOS / firmware
PROM / EPROM / EEPROM

Primary memory: RAM vs ROM.

LEARN IT

Primary / main memory sits on the motherboard and is directly used by the CPU. RAM is volatile: its contents vanish when power is cut. ROM is non-volatile: it keeps BIOS even when the PC is off.

RAM	ROM
Random Access Memory	Read Only Memory
Read and write	Mostly read; writing needs special process
Volatile	Non-volatile
Stores OS + running programs + data	Stores firmware / BIOS
Larger, cheaper per bit (DRAM)	Smaller

RAM types

- SRAM (Static): uses flip-flops, faster, used as cache, more expensive, no refresh.
- DRAM (Dynamic): uses capacitors, needs refresh, cheaper, used as main memory.

- SIMM = Single Inline Memory Module (board MCQ).

ROM types

- PROM: programmed once by the user.
- EPROM: erased with ultraviolet light, then reprogrammed.
- EEPROM: electrically erased and reprogrammed, used for modern BIOS.

HOW BIEK ASKS THIS

MCQ: 1 GB = 1024 MB. Never write 1000. Difference of primary vs secondary: speed, volatility, capacity, CPU distance, cost.

COMMON BOARD MISTAKE

1 GB is 1024 MB, not 1024 KB. Cache is high-speed SRAM inside or near the CPU, not the hard disk.

اردو میں سمجھیں

سب سے چھوٹی اکائی بٹ ہے (0 یا 1)۔ آٹھ بٹ کا بائٹ ایک حرف رکھتا ہے۔ 1024 بائٹ = 1 KB۔
RAM عارضی ہے، بجلی جانے پر خالی۔ ROM مستقل ہے، BIOS رکھتی ہے۔ SRAM تیز کیش ہے، DRAM مین میموری ہے۔

CHECK YOURSELF

1. Convert: 2 KB = ? bytes
2. Which ROM is erased by UV light?

Answers: 1. $2 \times 1024 = 2048$ bytes · 2. EPROM

MCQs

1. 1 Gigabyte is equivalent to:
 - a. 1024 KB
 - b. 1024 Bytes
 - c. 1024 MB
 - d. 1024 TB
2. SIMM is the abbreviation of:
 - a. Single interfacing Memory Module
 - b. Simple Inline Memory Module
 - c. Single Inline Memory Module
 - d. Simple Inline Memory Mode
3. Smallest unit of data is:
 - a. Bit
 - b. Byte
 - c. Megabyte
 - d. Nibble

Key: 1=c, 2=c, 3=a

Short questions (3 marks each)

1. Differentiate RAM and ROM.
2. Explain SRAM and DRAM.

3. Write memory measurement units from bit to TB.

Model short answers

1. RAM: volatile R/W working memory. ROM: non-volatile firmware memory. RAM loses data without power; ROM does not.
2. SRAM is fast cache without refresh. DRAM is cheaper main memory that must be refreshed.
3. Bit, nibble (4 bits), byte (8 bits), KB, MB, GB, TB — each step $\times 1024$ except bit/nibble/byte.

Long questions (board Section C)

1. What is computer memory? Explain primary memory and its types with a comparison table.

Secondary storage and memory devices ★ Golden

One college period · 40–45 minutes · Computer Memory

What you will learn

- Explain magnetic, optical and solid-state secondary storage
- Give examples: HDD, USB, SD, CD/DVD/Blu-ray, external drive
- Compare primary memory with secondary storage for a long answer

Secondary memory. Non-volatile storage used to keep programs and data permanently, cheaper and larger than RAM but slower.

Family	How it stores	Examples
Magnetic	Magnetised spots on spinning platters / tape	Hard disk, floppy (old), magnetic tape
Optical	Laser pits on disc	CD, DVD, Blu-ray
Solid state / flash	Electric charge in chips, no moving parts	USB flash, SSD, SD / MicroSD

- Hard disk drive (HDD): large capacity, inside the system unit, moving heads — slower than SSD.
- SSD: flash chips, silent, fast boot, used in modern laptops.
- USB / pen drive / data traveler: portable flash memory.
- SD / MicroSD: phones and cameras.
- Combo drive: CD+DVD reader/writer in one bay.
- External HDD: backup connected by USB.

Primary memory	Secondary memory
On motherboard, used by CPU directly	On disk / USB, used via I/O
Fast, expensive, smaller	Slow, cheap, huge
RAM is volatile	Non-volatile
Measured in GB today	Measured in GB–TB

HOW BIEK ASKS THIS

2026 Section C: “Explain secondary storage devices with examples.” Write magnetic, optical, solid-state with two examples each and one use.

اردو میں سمجھیں

ثانوی میموری مستقل ذخیرہ ہے: ہارڈ ڈسک، یو ایس بی، سی ڈی/ڈی وی ڈی، ایس ڈی کارڈ۔ یہ RAM سے سست لیکن سستی اور بڑی ہے۔

CHECK YOURSELF

1. Why is a USB called solid-state storage?
2. Which is faster for Windows boot: HDD or SSD?

Answers: 1. No moving parts; data in flash chips · 2. SSD

MCQs

1. A USB flash drive is an example of:
 - a. Optical storage
 - b. Magnetic tape
 - c. Solid-state storage
 - d. Cache
2. Blu-ray is a type of:
 - a. Magnetic disk
 - b. Optical disc
 - c. SRAM
 - d. Register

Key: 1=c, 2=b

Short questions (3 marks each)

1. Explain secondary storage devices with examples.
2. Differentiate primary and secondary memory.

Model short answers

1. Secondary storage is permanent: HDD/SSD (magnetic/solid state), CD/DVD/Blu-ray (optical), USB and SD (flash). Used for files, OS install and backup.
2. Primary is fast, CPU-close, costly, RAM volatile. Secondary is slow, cheap, large, non-volatile.

Long questions (board Section C)

1. Explain secondary storage devices with examples. Compare them with primary memory.

System unit, motherboard, ports and expansion slots

One college period · 40–45 minutes · Inside System Unit

What you will learn

- Describe the system unit / CPU casing and its main parts
- Label motherboard components: CPU socket, RAM slots, BIOS, CMOS battery, buses, ports, expansion slots
- List serial, parallel, USB and HDMI ports
- State PCI / expansion slot purpose

System unit. The case that houses the motherboard, power supply, drives, cooling fans and ports — the “CPU box” of a desktop.

- Power supply (SMPS): converts 220 V AC to DC for components.
- Motherboard / main board: printed circuit that connects everything.
- Hard drive / combo drive bays.
- Ports on the back/front panel.
- Cooling: heat sink + fan, sometimes liquid cooling.

Motherboard components

Part	Function
Microprocessor socket	Holds the CPU
RAM slots	DIMM / SIMM modules of main memory
BIOS chip (ROM)	Starts the computer, POST, finds boot device
CMOS battery	Keeps date, time and BIOS settings when PC is off
Chipset	Northbridge/southbridge traffic between CPU, memory, I/O
Jumpers	Tiny connectors that set hardware options
Expansion slots	PCI / PCIe cards: graphics, Wi-Fi, sound
System bus	Electrical pathways for address, data, control

Ports

Port	Use
Serial (RS-232, old)	One bit at a time; mouse/modem in old PCs
Parallel (LPT)	Several bits at a time; old printers
USB	Universal, hot-pluggable: keyboard, disk, phone
HDMI	High-Definition Multimedia Interface — video+audio to TV/monitor
VGA	Analog video (older monitors)

REMEMBER IT

CMOS battery dead $\Rightarrow$ date/time reset every boot, sometimes "disk boot failure" until BIOS is set again.

HOW BIEK ASKS THIS

Short: components of the system unit. Long: motherboard with a sketch of CPU, RAM, slots, ports.

اردو میں سمجھیں

سسٹم یونٹ ڈیسک ٹاپ کا باکس ہے۔ مدر بورڈ سب کو جوڑتی ہے۔ پاور سپلائی بجلی دیتی ہے، پورٹس کی بورڈ اور مانیٹر لگاتے ہیں، USB عام پورٹ ہے۔ CMOS بیٹری تاریخ/وقت رکھتی ہے۔

CHECK YOURSELF

1. Which port carries video and audio together?
2. What does BIOS do at startup?

Answers: 1. HDMI · 2. POST and locate a bootable disk / OS

MCQs

1. The electrical path that carries data from one device to another is called:
 - a. Bus
 - b. CPU
 - c. Circuit only
 - d. Chip
2. USB stands for:
 - a. Universal Serial Bus
 - b. United System Board
 - c. Ultra Speed Bit
 - d. Uniform Serial Box

Key: 1=a, 2=a

Short questions (3 marks each)

1. Write the main components of a system unit.
2. What is CMOS battery used for?
3. Differentiate serial and USB ports.

Model short answers

1. Power supply, motherboard, CPU, RAM, storage drives, cooling, ports, expansion cards.
2. It powers CMOS memory so BIOS settings, date and time survive when the PC is unplugged.
3. Serial (old) sends one bit at a time; USB is a modern serial bus that is fast, hot-pluggable and powers devices.

Long questions (board Section C)

1. Describe the motherboard. Explain its components with a labelled diagram.

CPU, registers, buses and the fetch cycle ★ Golden

One college period · 40–45 minutes · Inside System Unit

What you will learn

- Describe ALU, CU, registers and cache
- Explain types of registers used in board answers
- Differentiate address, data and control buses
- Draw and explain the fetch–decode–execute cycle

Microprocessor / CPU. The integrated circuit that fetches instructions from memory, decodes them and executes them. It is the brain of the computer.

Part	Role
ALU (Arithmetic and Logic Unit)	Add, subtract, AND, OR, NOT, compare
CU (Control Unit)	Directs the flow of data; sends control signals
Registers	Very fast storage inside the CPU for one instruction/data word
Cache	Small SRAM holding recently used instructions/data
Clock	Pulses that time every step (GHz)

Registers (board favourite)

- ACC (Accumulator): stores ALU results.
- PC (Program Counter): address of the next instruction.
- IR (Instruction Register): the instruction currently being decoded.
- MAR (Memory Address Register): address being accessed in RAM.
- MDR (Memory Data Register): data coming from or going to RAM.
- General purpose registers: AX, BX, CX, DX in old x86 teaching notes.
- Flag register: zero, carry, sign bits after ALU operations.

Buses

Bus	Carries	Direction
Address bus	Memory / I/O addresses	Unidirectional (CPU → memory)
Data bus	Actual data and instructions	Bidirectional
Control bus	Read, write, interrupt, clock	Mostly CPU → others (some lines back)



repeats for the next instruction

Instruction cycle. Board wording: Fetch Cycle (often they mean the whole fetch-execute cycle).

LEARN IT

Fetch: PC gives address, instruction copied from memory into IR, PC incremented. Decode: CU interprets the opcode. Execute: ALU or memory operation happens. Store/write-back: result goes to a register or RAM. Then the cycle repeats.

HOW BIEK ASKS THIS

2026 Q2(iv): "What is fetch Cycle? Draw its diagram." Draw four boxes Fetch→Decode→Execute→Store with a loop arrow. Q3 OR: Computer Register and its types.

REMEMBER IT

Clock speed, word size (32/64 bit), cache size and number of cores decide CPU power. Generation: 1st vacuum tubes ... 3rd ICs ... 4th microprocessors ... 5th AI.

اردو میں سمجھیں

CPU میں ALU حساب کرتی ہے، CU حکم دیتی ہے، رجسٹر بہت تیز چھوٹی میموری ہیں۔
Fetch سائیکل: میموری سے ہدایت لاؤ، سمجھو، چلاؤ، نتیجہ رکھو۔ ایڈریس بس پتہ لے جاتی ہے، ڈیٹا بس ڈیٹا، کنٹرول بس اشارے۔

CHECK YOURSELF

1. Which register holds the next instruction address?
2. Is the data bus uni- or bi-directional?

Answers: 1. Program Counter (PC) · 2. Bidirectional

MCQs

1. The third generation of computers used:
 - a. Vacuum tubes
 - b. Transistors
 - c. Integrated circuits
 - d. Microprocessors
2. ALU is a part of the:
 - a. Printer
 - b. CPU
 - c. Monitor

d. Keyboard

Key: 1=c, 2=b

Short questions (3 marks each)

1. What is the fetch cycle? Draw its diagram.
2. What is a computer register? Explain its types.
3. Differentiate address bus and data bus.

Model short answers

1. The CPU repeatedly fetches an instruction from memory, decodes it, executes it and stores the result. Diagram: Fetch → Decode → Execute → Store, looping.
2. A register is a small high-speed memory cell in the CPU. Types: accumulator, PC, IR, MAR, MDR, flags, general purpose.
3. Address bus sends locations one way from CPU; data bus carries the bits both ways.

Long questions (board Section C)

1. Explain the internal architecture of a microprocessor. Draw and describe the fetch-execute cycle.

Operating system: types, features and examples ★ Golden

One college period · 40–45 minutes · Operating System

What you will learn

- Define OS and its objectives
- Name Windows, macOS, DOS, UNIX/Linux, Android
- Explain batch, multi-tasking, time-sharing, multi-processing, distributed, embedded
- State that a computer cannot boot without an OS

Operating System. System software that manages hardware and software resources and provides an interface between the user and the computer. It is loaded at boot time. A PC cannot boot without an OS.

Common OS: Microsoft Windows, Apple macOS, DOS (old command-line), UNIX and Linux (open source), Android and iOS for mobiles.

Feature	Meaning	Example
Batch processing	Jobs queued; no user interaction while running	Old payroll runs, mainframes
Multi-tasking	Several programs appear to run at once	Browser + Word + music
Time sharing	CPU time sliced among users/processes	University server
Multi-processing	Two or more CPUs / cores share work	Dual-core laptop
Parallel processing	One job split across processors	Scientific calculation
Distributed	Many computers act as one system	Google data centres
Embedded	OS baked into a device	ATM, microwave, car ECU, Android phone

CLI (Command Line)	GUI (Graphical User Interface)
Type commands (DIR, CD)	Windows, icons, mouse
Fast for experts, low memory	Easy for beginners
DOS, UNIX shell	Windows, GNOME, macOS

HOW BIEK ASKS THIS

Long 2026: "What is an Operating System? Explain its Functions." Functions are the next lecture — keep definition + types here and functions ready.

COMMON BOARD MISTAKE

Oracle is a database, not an OS. Chrome is a browser (application). Linux is an OS.

آپریٹنگ سسٹم وہ سافٹ ویئر ہے جو کمپیوٹر کو بوٹ کرتا ہے، ہارڈویئر سنبھالتا ہے اور صارف کو انٹرفیس دیتا ہے۔ ونڈوز، لینکس، اینڈرائیڈ مشہور OS ہیں۔
ملٹی ٹاسکنگ کئی پروگرام ایک ساتھ؛ ایمبیڈڈ OS فریج یا موبائل کے اندر ہوتا ہے۔

CHECK YOURSELF

1. Can a computer boot without an OS?
2. Give one embedded OS example.

Answers: 1. No · 2. Android in a phone / OS in an ATM

MCQs

1. A computer cannot boot if it does not have a:
 - a. Compiler
 - b. Browser
 - c. Operating System
 - d. Loader only
2. Which is NOT a type of operating system?
 - a. Windows
 - b. Linux
 - c. Mac OS
 - d. Oracle
3. Windows is:
 - a. Application software
 - b. System software
 - c. A compiler
 - d. Firmware only

Key: 1=c, 2=d, 3=b

Short questions (3 marks each)

1. Define operating system. Name four operating systems.
2. Differentiate multi-tasking and multi-processing.
3. What is an embedded operating system?

Model short answers

1. OS is system software that manages resources and lets the user run programs. Windows, Linux, macOS, Android.
2. Multi-tasking: many programs on one CPU by time-slicing. Multi-processing: more than one processor executing at the same time.
3. An OS built into a dedicated device (router, car, TV, mobile) with a specific job.

Long questions (board Section C)

1. What is an operating system? Explain its types / features with examples.

Functions of OS, process management and GUI ★ Golden

One college period · 40–45 minutes · Operating System

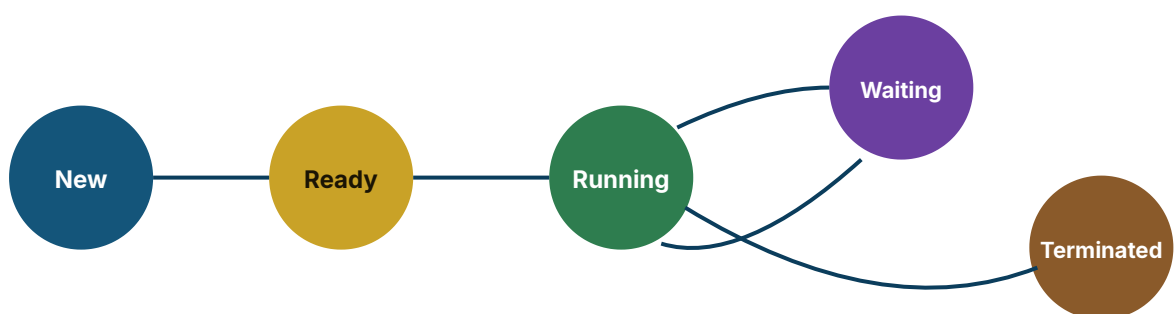
What you will learn

- List and explain the functions of an OS
- Define process and thread; draw process states
- Use files, folders, Control Panel, msconfig, dxdiag at a theory level
- Mention open-source OS and mobile OS

Functions of an operating system

Function	What the OS does
Booting	POST, load kernel, start system services
Process management	Create, schedule, suspend, terminate processes
Memory management	Allocate RAM, virtual memory / paging
File management	Create, copy, delete, rename files and folders; FAT/NTFS
I/O management	Talk to keyboard, disk, printer via drivers
Secondary storage management	Free space, disk scheduling
Network management	TCP/IP, sharing, firewall basics
Protection / security	Passwords, permissions, user accounts
Command interpreter / shell	cmd, PowerShell, Bash, or GUI Explorer

Process. A program in execution. A thread is a lightweight unit of a process that shares the same memory space. Multi-threading: many threads in one process. Multi-tasking: many processes.



Process states: New → Ready → Running → Waiting or Terminated.

- New: being created.
- Ready: waiting for CPU.
- Running: on the CPU.
- Waiting / blocked: waiting for I/O or an event.
- Terminated: finished or killed.

Working with a GUI OS (Windows)

Create / delete / copy / rename files; search; attributes (Read-only, Hidden, Archive, System); drives and paths (C:\Users\Ali\Notes.txt); Control Panel / Settings for hardware; dxdiag to list display and sound; msconfig to change startup programs.

Open-source OS (Linux, Ubuntu) can be studied, copied and improved freely. Mobile OS: Android versions, iOS.

HOW BIEK ASKS THIS

Long answer: OS definition + at least 7 functions with one line each. Short: process vs thread; five process states.

اردو میں سمجھیں

OS بوٹ کرتا ہے، پروسیس اور میموری بانٹتا ہے، فائلیں سنبھالتا ہے، پرنٹر چلاتا ہے اور پاس ورڈ سے حفاظت کرتا ہے۔
پروسیس چلتا ہوا پروگرام ہے۔ حالتیں: نیا، تیار، چل رہا، انتظار، ختم۔

CHECK YOURSELF

1. Name five OS functions.
2. Ready vs running?

Answers: 1. Booting, process, memory, file, I/O (any five) · 2. Ready waits for CPU; running currently has the CPU

MCQs

1. A program in execution is called a:
 - a. File
 - b. Process
 - c. Folder
 - d. Port
2. File management is a function of the:
 - a. Compiler
 - b. Operating system
 - c. Browser
 - d. ALU

Key: 1=b, 2=b

Short questions (3 marks each)

1. Explain the functions of an operating system.
2. Differentiate process and thread.
3. Write the states of a process.

Model short answers

1. Booting, process, memory, file, I/O, storage, network, protection, command interpreter — each in one sentence as in the table.
2. Process: whole program in execution with its own memory. Thread: a path inside a process sharing memory; lighter to create.
3. New, Ready, Running, Waiting/Blocked, Terminated.

Long questions (board Section C)

1. What is an Operating System? Explain its functions. Also describe process management.

Data communication: components, signals and modes ★ Golden

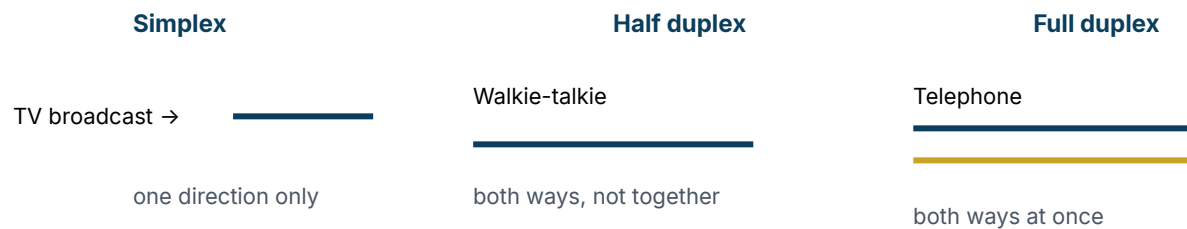
One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Define data communication and its five components
- Differentiate analog and digital signals
- Explain simplex, half duplex and full duplex
- Define modulation, demodulation, synchronous and asynchronous transmission

Data communication. The exchange of data between two devices through a transmission medium according to a protocol.

Five components (board list): Sender, Receiver, Message, Medium, Protocol.



Direction of data flow.

Mode	Direction	Example
Simplex	One way only	Radio, keyboard → CPU, TV broadcast
Half duplex	Both ways, not at the same time	Walkie-talkie
Full duplex	Both ways at the same time	Telephone, mobile call

Analog vs digital

Analog signal	Digital signal
Continuous wave, infinite values	Discrete 0 and 1 pulses
Human voice, old telephone line	Computers, optical fibre (light pulses)
Noise distorts easily	More reliable, regenerable

Modulation. Converting a digital signal into analog so it can travel on a telephone line (ASK/FSK/PSK). Performed by a modem on send.

Demodulation. Converting the analog signal back to digital at the receiving modem.

MODEM = MODulator + DEModulator.

Asynchronous	Synchronous
Start and stop bits around each character	A timing clock; data sent in blocks/frames
Cheaper, slower, keyboard typing	Faster, used on networks

Broadcast: one sender, many receivers (TV, radio). Point-to-point: one sender, one receiver (phone call).

HOW BIEK ASKS THIS

2026 shorts: analog vs digital; modulation/demodulation; broadcast vs point-to-point; components of data communication; half duplex MCQ.

COMMON BOARD MISTAKE

Full duplex is both directions simultaneously. Half duplex is both directions but one at a time. Students mix these every year.

اردو میں سمجھیں

ڈیٹا کمیونیکیشن میں بھیجنے والا، وصول کرنے والا، پیغام، میڈیم اور پروٹوکول ہوتے ہیں۔
Simplex ایک طرف، Half duplex باری باری دونوں طرف، Full duplex ایک ساتھ دونوں طرف۔ Modulation ڈیجیٹل کو
اینالوگ بناتا ہے، Demodulation الٹ۔

CHECK YOURSELF

1. Telephone is which mode?
2. Modem expands to?

Answers: 1. Full duplex · 2. Modulator Demodulator

MCQs

1. A mode that allows two directions but not simultaneously is:
 - a. Full duplex
 - b. Half duplex
 - c. Synchronous
 - d. Simplex
2. Converting analog to digital is called:
 - a. Modulation
 - b. Telecommunicating
 - c. Demodulation
 - d. Switching
3. Protocol means:
 - a. A topology
 - b. A set of rules
 - c. A router
 - d. A modem

Key: 1=b, 2=c, 3=b

Short questions (3 marks each)

1. Name the components of data communication.
2. Differentiate analog and digital signals.
3. What is modulation and demodulation?
4. Differentiate broadcast and point-to-point.
5. Differentiate synchronous and asynchronous transmission.

Model short answers

1. Sender, receiver, message, medium, protocol.
2. Analog is continuous; digital is 0/1. Digital resists noise better.
3. Modulation: digital→analog for the line. Demodulation: analog→digital. Modem does both.
4. Broadcast: one-to-many. Point-to-point: one-to-one.
5. Async uses start/stop bits per character. Sync uses a clock and sends blocks — faster.

Long questions (board Section C)

1. Explain data communication. Describe communication modes and analog/digital signals with diagrams.

Transmission media and network devices ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Classify guided and unguided media
- Describe twisted pair, coaxial and fibre optic
- Describe radio, microwave and infrared
- Explain NIC, modem, hub, switch, router, bridge, gateway

Guided (wired) media

Medium	Construction	Use / note
Twisted pair	Pairs of copper wires twisted (UTP/STP)	LAN cables Cat5/Cat6; cheap
Coaxial	Core, insulation, metal shield, jacket	Old TV cable, some backbone
Fibre optic	Glass/plastic, light pulses	Very fast, long distance, no EMI

Unguided (wireless) media

- Radio waves: FM radio, Wi-Fi, Bluetooth, cellular.
- Microwaves: line-of-sight dishes, some mobile backhaul.
- Infrared: TV remote, old short laptop links; blocked by walls.

LAN / communication devices

Device	Job
NIC (Network Interface Card)	Fits in PC; has MAC address
Hub	Repeats bits to all ports (old, collisions)
Switch	Forwards frames to the correct port using MAC
Bridge	Connects two similar LANs / segments
Router	Connects different networks using IP; path to the internet
Gateway	Connects two different networks/protocols (e.g. LAN to mainframe)
Modem	Analog↔digital for the phone/cable line
Access point	Wi-Fi radio for wireless clients

HOW BIEK ASKS THIS

2026: coaxial and fibre are communication media; bridge connects two similar networks; router/gateway for different networks; SMTP is the common email protocol.

REMEMBER IT

Memory: Switch is smart hub. Router is smart switch that knows IP. Gateway is a protocol translator. Bridge = two similar LANs.

گائیڈڈ میڈیم تار ہے: ٹوئسٹڈ پیئر، کواکسیئل، فائبر آپٹک۔ ان گائیڈڈ ریڈیو، مائیکرو ویو، انفراریڈ۔

سوئچ LAN کے کمپیوٹر جوڑتا ہے، روٹر مختلف نیٹ ورک اور انٹرنیٹ، بریج دو ملتے جلتے LAN، گیٹ وے مختلف پروٹوکول۔

CHECK YOURSELF

1. Which cable uses light?
2. Which device uses IP addresses to connect networks?

Answers: 1. Fibre optic · 2. Router

MCQs

1. Coaxial cable and fibre optics are examples of:
 - a. Router
 - b. Communication media
 - c. Modem
 - d. Switch
2. The LAN component that connects two similar networks is:
 - a. Server
 - b. Bridge
 - c. Router
 - d. Gateway
3. The most common protocol used for e-mail is:
 - a. FTP
 - b. TCP/IP
 - c. SMTP
 - d. IEEE

Key: 1=b, 2=b, 3=c

Short questions (3 marks each)

1. Explain data communication channels / transmission media with examples.
2. Define the LAN component which allows communication between two different networks.
3. Differentiate hub and switch.

Model short answers

1. Guided: twisted pair, coaxial, fibre. Unguided: radio, microwave, infrared. Fibre is fastest and immune to electrical noise.
2. Router or gateway. A router forwards IP packets between networks; a gateway also converts protocols.
3. Hub broadcasts to all ports; switch sends a frame only to the destination MAC port.

Long questions (board Section C)

1. Explain data communication channels with examples. Describe common networking devices.

Networks, topologies and architecture ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

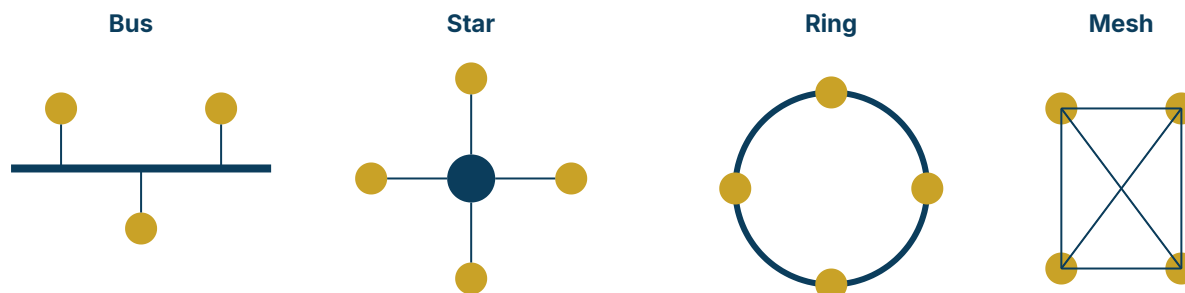
What you will learn

- Define network and its advantages
- Classify LAN, MAN, WAN
- Draw bus, star, ring, mesh, tree, hybrid topologies
- Name ISO, IEEE, ITU; introduce OSI and TCP/IP

Computer network. Two or more computers connected to share resources (files, printers, internet) and communicate.

Advantages: resource sharing, cheaper software/hardware per user, easy communication, central backup. Disadvantages: security risk, server failure, cabling cost, viruses spread faster.

Type	Span	Example
LAN	Room, lab, building	College computer lab
MAN	City	Cable TV / city Wi-Fi backbone
WAN	Country / world	Internet, bank branches



Four topologies every Section C diagram should match.

Topology	How linked	Plus / minus
Bus	One backbone cable	Cheapest; cable break kills the LAN; BIEK MCQ: cheapest topology
Star	Each node to a hub/switch	Easy to add PCs; centre fails ⇒ all fail
Ring	Circle, token passing (old)	Equal access; one break can stop ring
Mesh	Many-to-many	Very reliable; expensive
Tree	Stars of stars	Scalable campus
Hybrid	Mix	Real offices

Standard bodies: ISO (OSI model), IEEE (802.3 Ethernet, 802.11 Wi-Fi), ITU (telecom), ASCII (character codes — often listed with them in the Sindh book).

HOW BIEK ASKS THIS

Long 2026 OR: "What is meant by Computer Network? Classify by their scale and limitation." Write LAN/MAN/WAN with distance and one limitation each, then topologies with diagrams.

اردو میں سمجھیں

نیٹ ورک دو یا زیادہ کمپیوٹر جو وسائل بانٹتے ہیں۔ LAN عمارت، MAN شہر، WAN دنیا (انٹرنیٹ)۔
بس سب سے سستی، اسٹار سوئچ پر، رینگ دائرہ، میش ہر ایک سے ہر ایک۔

CHECK YOURSELF

1. Internet is which scale of network?
2. Cheapest topology?

Answers: 1. WAN · 2. Bus

MCQs

1. The cheapest topology is:
 - a. Ring
 - b. Bus
 - c. Tree
 - d. Star
2. OSI model contains:
 - a. six layers
 - b. five layers
 - c. three layers
 - d. seven layers
3. Which is NOT a type of computer network?
 - a. LAN
 - b. MAN
 - c. WAN
 - d. CPU

Key: 1=b, 2=d, 3=d

Short questions (3 marks each)

1. What is a computer network? Give advantages.
2. Differentiate LAN and WAN.
3. Write short notes on star and bus topology.

Model short answers

1. Linked computers sharing resources. Advantages: sharing, communication, backup, cost.
2. LAN is local and fast (lab). WAN covers large distance (internet) and is slower/more expensive.
3. Star: nodes to central switch, easy management. Bus: shared cable, cheap, collision and break problems.

Long questions (board Section C)

1. What is meant by a computer network? Classify by scale and explain topologies with diagrams.

OSI reference model, TCP/IP and internet protocols ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Draw seven OSI layers with one function and one protocol/device each
- Map TCP/IP four layers onto OSI
- Define protocol, TCP, IP, HTTP, FTP, SMTP, DNS, URL

Protocol. A set of rules that sender and receiver follow so communication is understood (format, timing, error handling).

7 Application	HTTP, SMTP, FTP, DNS
6 Presentation	encryption, compression, translation
5 Session	start, manage, end sessions
4 Transport	TCP, UDP — ports, reliability
3 Network	IP addressing, routing
2 Data Link	frames, MAC, switches
1 Physical	cables, bits, hubs, NICs

ISO OSI 7-layer model — learn bottom to top: Please Do Not Throw Sausage Pizza Away (Physical → Application).

LEARN IT

Physical: bits on wire. Data link: frames, MAC, switch. Network: packets, IP, router. Transport: segments, TCP (reliable) / UDP (fast). Session: connections. Presentation: encrypt/compress. Application: user programs.

TCP/IP layer	OSI layers	Examples
Application	5+6+7	HTTP, SMTP, FTP, DNS
Transport	4	TCP, UDP
Internet	3	IP, ICMP
Network access / Link	1+2	Ethernet, Wi-Fi

Protocol	Use
TCP/IP	Suite that runs the internet
HTTP / HTTPS	Web pages (S = encrypted)
FTP	File transfer
SMTP	Sending email
POP3 / IMAP	Receiving email
DNS	Name → IP (www.biek.edu.pk)
URL	Uniform Resource Locator — web address

WORKED EXAMPLE / BOARD STYLE

URL example: <https://www.biek.edu.pk/index.html> — protocol https, host www.biek.edu.pk, path /index.html.

HOW BIEK ASKS THIS

2026 long: Draw and explain OSI. Also MCQ: seven layers; SMTP for email. Write one sentence per layer and a small stack diagram.

اردو میں سمجھیں

OSI ماڈل سات تہوں میں نیٹ ورک سمجھاتا ہے: فزیکل سے ایپلی کیشن تک۔ TCP/IP انٹرنیٹ کا عملی ماڈل ہے (چار تہیں)۔

پروٹوکول قواعد ہیں: SMTP ای میل بھیجنے، HTTP ویب، FTP فائل۔ URL ویب ایڈریس ہے۔

CHECK YOURSELF

1. How many OSI layers?
2. Which protocol sends email?

Answers: 1. Seven · 2. SMTP

MCQs

1. OSI model has:
 - a. 4 layers
 - b. 5 layers
 - c. 6 layers
 - d. 7 layers
2. URL stands for:
 - a. Universal Resource Locator
 - b. Uniform Resource Locator
 - c. Unique Resource Locator
 - d. Ultimate Resource Locator
3. HTTP works at the OSI:
 - a. Physical layer
 - b. Network layer
 - c. Application layer

d. Data link layer

Key: 1=d, 2=b, 3=c

Short questions (3 marks each)

1. Write the seven layers of the OSI model.
2. Differentiate TCP/IP and OSI.
3. What is a protocol? Give three examples.

Model short answers

1. Physical, Data Link, Network, Transport, Session, Presentation, Application — with one function each.
2. OSI is a 7-layer ISO reference model. TCP/IP is the 4-layer practical internet model. TCP/IP application covers OSI 5–7.
3. Rules of communication. TCP, IP, HTTP, SMTP, FTP.

Long questions (board Section C)

1. Draw and explain the OSI reference model. Give protocols at each layer.

Number sense, generations, abbreviations and security recap

One college period · 40–45 minutes · Paper I mixed theory

What you will learn

- Recall computer generations
- Expand the abbreviations that appear every year
- Restate virus / antivirus and basic security
- Keep a one-page formula sheet for Paper I

Generation	Technology	Example / note
1st (1940s–50s)	Vacuum tubes	ENIAC, huge, hot
2nd	Transistors	Smaller, more reliable
3rd	Integrated circuits (ICs)	Board MCQ favourite
4th	Microprocessor	IBM PC, personal computers
5th	AI, ULSI, natural language	Modern research / assistants

Full forms to memorise

Abbrev.	Full form
CPU	Central Processing Unit
ALU / CU	Arithmetic and Logic Unit / Control Unit
RAM / ROM / BIOS	Random Access Memory / Read Only Memory / Basic Input Output System
USB / HDMI / VGA	Universal Serial Bus / High-Definition Multimedia Interface / Video Graphics Array
OCR / OMR / MICR / CRT	Optical Character / Mark / Magnetic Ink Character Recognition / Cathode Ray Tube
GUI / CLI / OS	Graphical User Interface / Command Line Interface / Operating System
LAN / MAN / WAN / NIC	Local / Metropolitan / Wide Area Network / Network Interface Card
OSI / ISO / IEEE / SMTP / FTP / HTTP / DNS / URL / IP / TCP	Open Systems Interconnection / International Organization for Standardization / Institute of Electrical and Electronics Engineers / Simple Mail Transfer Protocol / File Transfer Protocol / HyperText Transfer Protocol / Domain Name System / Uniform Resource Locator / Internet Protocol / Transmission Control Protocol
SIMM / SRAM / DRAM / EEPROM	Single Inline Memory Module / Static / Dynamic RAM / Electrically Erasable PROM
AI / ICT / IT	Artificial Intelligence / Information and Communication Technology / Information Technology

REMEMBER IT

Every night before the paper, write this table from memory. Section B often says “write full form of any three”.

CHECK YOURSELF

1. 3rd generation technology?
2. BIOS full form?

Answers: 1. Integrated circuits · 2. Basic Input Output System

MCQs

1. OCR stands for:
 - a. Optical Character Recognition
 - b. Original Code Reader
 - c. Output Card Reader
 - d. Open Computer RAM
2. GUI stands for:
 - a. General User Internet
 - b. Graphical User Interface
 - c. Global Unique ID
 - d. General Utility Input

Key: 1=a, 2=b

Short questions (3 marks each)

1. Write full forms of OCR, GUI, BIOS, SMTP, USB.
2. Write a short note on computer generations.

Model short answers

1. OCR Optical Character Recognition; GUI Graphical User Interface; BIOS Basic Input Output System; SMTP Simple Mail Transfer Protocol; USB Universal Serial Bus.
2. Five generations from vacuum tubes to microprocessors and AI, each smaller, faster and cheaper.

Programming concepts that start in XI and are examined in XII

One college period · 40–45 minutes · Programming bridge (Sindh XI C++)

What you will learn

- Know that BIEK examines C in Paper II, while the Sindh 2019 book uses C++ in XI
- Recall algorithm, flowchart, source vs object code, IDE
- Preview data types, control structures, arrays and functions

COMMON BOARD MISTAKE

Karachi Board Paper I (XI) is theory. Full C programs are Paper II. Still, Sindh Unit 5–6 expect you to practise C++ in the lab in Year 1. This lecture is the bridge; Class XII lectures teach C in exam style.

Algorithm. A step-by-step finite method to solve a problem. Pseudocode is the algorithm in English-like lines. A flowchart uses standard symbols (terminal, process, input/output, decision).

IDE. Integrated Development Environment — editor + compiler + debugger in one window (Turbo C++, Dev-C++, VS Code). Shortcuts in Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 user screen.

C/C++ program structure: preprocessor (#include), global declarations, main(), statements, return. Tokens: keywords, identifiers, constants, operators, separators. C is case-sensitive: int is a keyword, Int is not.

Idea	C / C++ sketch
Input / output	scanf("%d", &n); printf("%d", n); or cin >> n; cout << n;
Selection	if, if-else, else-if, switch-case, nested if
Loop	for, while, do-while, nested loops; break, continue, goto
Function	return-type name(parameters) { body }
Array	int a[5] = {1,2,3,4,5};
String	char s[20] = "BIEK"; strlen, strcpy, strcat, strcmp
Structure	struct Employee { char name[20]; float salary; };

```
#include <stdio.h>
int main(void) {
    int m1 = 80, m2 = 70, m3 = 90;
    float avg = (m1 + m2 + m3) / 3.0f;
    if (avg ≥ 40) printf("Pass %.2f\n", avg);
    else printf("Fail\n");
    return 0;
}
```

Lab list from the Sindh curriculum (XI): calculator, marksheet, prime test, digits of a number, factors, star pattern, larger of two via function, sort array, search, matrix add/multiply, reverse a name with strlen, employee structure.

الخوارزم مسئلہ حل کرنے کے قدم ہیں۔ فلو چارٹ خاکہ ہے۔ IDE میں کوڈ لکھ کر چلاتے ہیں۔ BIEK کے پیپر نو میں C زبان آتی ہے — اگلی کتاب (XII) میں پوری مشق ہے۔

CHECK YOURSELF

1. F9 in Turbo C?

2. Is C case-sensitive?

Answers: 1. Compile · 2. Yes

MCQs

1. An IDE is used to:
 - a. Print books
 - b. Write and run programs
 - c. Scan photos only
 - d. Connect fibre
2. Which is a loop structure?
 - a. if
 - b. switch
 - c. for
 - d. goto only

Key: 1=b, 2=c

Short questions (3 marks each)

1. What is an IDE? Write a few shortcut keys.
2. Differentiate source code and object code.
3. Why is C called case-sensitive?

Model short answers

1. IDE combines editor, compiler and debugger. Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 output screen.
2. Source code is human-written (file.c). Object/machine code is the compiler output that the CPU runs.
3. Uppercase and lowercase letters are different: TOTAL and total are two identifiers; keywords must be lowercase (int, not INT).

Paper I mock (BIEK 2026 style) with model answers ★ Golden

One college period · 40–45 minutes · Revision

What you will learn

- Attempt a full Paper I in one sitting
- Match answers to the marking style of Section B and C

Section A — 15 MCQs (key at the end)

- 1) 1 GB = ? (a) 1024 KB (b) 1024 B (c) 1024 MB (d) 1024 TB
- 2) Floppy and hard disk are (a) optical (b) magnetic tape (c) magnetic disk (d) flash
- 3) SIMM means (c) Single Inline Memory Module
- 4) OSI layers (d) seven
- 5) Cheapest topology (b) Bus
- 6) Virus is (b) software
- 7) Scanning input (d) OMR
- 8) Path for data between devices (a) Bus
- 9) Two similar networks (b) Bridge
- 10) Coaxial and fibre are (b) communication media
- 11) Web markup (b) HTML
- 12) Email send protocol (c) SMTP
- 13) Analog→digital (c) demodulation
- 14) Rules on the internet (b) protocol
- 15) Two ways not together (b) half duplex

Section B — write 6–8 lines each

1. IT: definition + 3 uses + 3 abuses OR barcode in a superstore.
2. Hardcopy vs softcopy OR sync vs async transmission.
3. Virus + 5 antiviruses OR plotter.
4. Fetch cycle + diagram OR five communication components.
5. Modulation/demodulation OR compiler vs interpreter.
6. Four differences: impact vs non-impact printers.
7. Device that joins two different networks (router/gateway).
8. Analog vs digital.
9. Broadcast vs point-to-point.
0. Full forms: OCR, GUI, SVGA, CRT, BIOS (any three).

Section C — diagrams compulsory

1. OS and its functions OR registers and types.
2. Communication channels with examples OR secondary storage.

3. OSI model OR network classification by scale + topologies.

MCQ key

1c 2c 3c 4d 5b 6b 7d 8a 9b 10b 11b 12c 13c 14b 15b

WORKED EXAMPLE / BOARD STYLE

Model long: OSI — Draw seven labelled rectangles. Write: Physical bits; Data link frames/MAC; Network IP/routing; Transport TCP ports; Session dialogues; Presentation encryption; Application HTTP/SMTP. Mention ISO. 8–10 marks if diagram + all layers are named.

اردو میں سمجھیں

ماک پیپر اسی انداز کا ہے جیسا 2026 ماڈل پیپر۔ پہلے MCQ، پھر مختصر، پھر خاکوں والے لمبے سوال۔ وقت تقسیم کریں: 20 منٹ MCQ، باقی دو حصے برابر۔

REMEMBER IT

On the real OMR: blue/black ballpoint only, no whitener. In the answer book, underline headings and box the final comparison table.

MCQs

1. Paper I mock key starts with 1 GB = 1024 MB. That option is:

- a. a
- b. b
- c. c
- d. d

Key: 1=c

Long questions (board Section C)

1. Attempt Section C question 3 of this mock under timed conditions and compare with the OSI model lecture.

BIEK Paper II: Option I (C) or Option II (Visual Basic)

One college period · 40–45 minutes · Paper II map

What you will learn

- Choose the option you studied in college — you cannot mix C and VB in one paper
- State that Database / MS Access is common to both options
- Know Section A 15 MCQs, Section B 10 of 15 shorts, Section C 3 of 5 longs (2026 model)

LEARN IT

Paper II is 75 marks. The front page says: attempt ONE option — Programming using C OR Programming using Visual Basic. Database questions (keys, MS Access objects, data models, SQL) appear in BOTH options.

	Option I — C	Option II — Visual Basic
Language	Turbo C / ANSI C in answer book	VB 6 style forms + code
Typical programs	factorial, table, greater of two, patterns, functions	even numbers, sum/average, tax slabs, calculator form
Theory	tokens, operators, loops, arrays, functions, pointers	controls, DIM, loops, arrays, built-in functions, errors
Shared	DBMS, tables, keys, ER, Access data types, SQL	same

Sindh Curriculum 2019 Grade XII also includes SDLC, pointers, OOP with C++, file handling, multimedia and wireless. Colleges that follow the new book still need C or VB for the BIEK paper, so this pack teaches both the board options and the extra Sindh units.

HOW BIEK ASKS THIS

2026 Option I shorts: structure of C, #include/#define, multiplication table, break vs continue, reserved words, address operator, factorial, scanf vs getchar, primary vs foreign key, expression values, DBMS components, function to print greater number, maths→C, nested-loop patterns, table/query/report.

COMMON BOARD MISTAKE

void is a reserved word, so it is an invalid identifier. Identifiers cannot start with a digit or contain a space: 1st number is invalid; std_marks is valid.

اردو میں سمجھیں

بارہویں کا پیپر ٹو دو آپشن میں ہے: C زبان یا Visual Basic. دونوں میں MS Access / ڈیٹا بیس آتا ہے۔ جو آپشن کالج میں پڑھا ہے وہی حل کریں۔

CHECK YOURSELF

1. Can you answer a C long question in the VB option?
2. Is MS Access only in Option II?

Answers: 1. No — pick one option for the whole paper · 2. No — database is in both options

MCQs

1. Every C program must contain:
 - a. start()
 - b. getch()
 - c. main()
 - d. printf()
2. Dennis Ritchie founded:
 - a. C
 - b. BASIC
 - c. JAVA
 - d. PASCAL
3. In MS Access a row is also called a:
 - a. Field
 - b. Column
 - c. Record
 - d. Query

Key: 1=c, 2=a, 3=c

C language, structure of a program and Turbo C IDE ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write the basic structure of a C program
- Explain #include and #define with examples
- Use Turbo C shortcuts and know compiler vs editor vs debugger
- List reserved words and rules of identifiers

C was designed by Dennis Ritchie at Bell Labs (1972) to write UNIX. It is a middle-level, compiled, case-sensitive language. BIEK still expects Turbo C style in many colleges: #include <stdio.h>, #include <conio.h>, void main(), clrscr(), getch().

```
#include <stdio.h>
#include <conio.h>
#define PI 3.14
void main()
{
    clrscr();
    printf("Board of Intermediate Education Karachi\n");
    getch();
}
```

Part	Job
#include <stdio.h>	Preprocessor: copies standard I/O functions (printf, scanf)
#include <conio.h>	Turbo C console: clrscr, getch
#define PI 3.14	Manifest constant — name replaced before compile
void main()	Execution starts here (ISO C prefers int main(void))
Declarations	Variables before use in C89 / Turbo C
Statements	End with semicolon
getch()	Wait for a key so the output screen stays

Identifiers and keywords

- Letters, digits, underscore; cannot start with a digit; no spaces or special symbols; cannot be a keyword.
- Valid: std_marks, NUMBER5, Number_5. Invalid: 5number, 1st number, void, int, float.
- Keywords (reserved): auto, break, case, char, const, continue, default, do, double, else, enum, extern, float, for, goto, if, int, long, register, return, short, signed, sizeof, static, struct, switch, typedef, union, unsigned, void, volatile, while.

Escape sequence. A backslash code inside a string: \n new line, \t tab, \\ backslash, \a beep, \" quote, \0 string terminator.

HOW BIEK ASKS THIS

2025/26 shorts: structure of C; #include or #define; reserved words; valid/invalid identifiers; why C is case sensitive; IDE shortcuts.

اردو میں سمجھیں

C پروگرام #include سے شروع ہوتا ہے، main() سے چلتا ہے۔ #define نام کو قدر سے بدل دیتا ہے۔ (int, Reserved words void, for نام نہیں بن سکتے۔)

CHECK YOURSELF

1. Who developed C?

2. What does \n do?

Answers: 1. Dennis Ritchie · 2. Moves the cursor to a new line

MCQs

1. C programs are converted to machine language by a:
 - a. Editor
 - b. Assembler
 - c. Compiler
 - d. Debugger
2. This is not a valid identifier:
 - a. Number
 - b. 5number
 - c. NUMBER5
 - d. Number_5
3. Comments in C are enclosed in:
 - a. /* ... */
 - b. /* ... /*
 - c. */ ... /*
 - d. ** **
4. Names of variables and functions are called:
 - a. header files
 - b. identifiers
 - c. loops
 - d. structures

Key: 1=c, 2=b, 3=a, 4=b

Short questions (3 marks each)

1. Write the basic structure of a C program.
2. Define #include or #define with two examples.
3. What is an IDE? Write a few shortcut keys.
4. Identify valid/invalid: void, std_marks, 1st number.
5. Why is C called case-sensitive? Define three reserved words.

Model short answers

1. Preprocessor directives, global declarations, main(), local declarations, statements, closing brace.
Show a 8-line skeleton as above.
2. #include <stdio.h> brings printf. #include <math.h> brings sqrt. #define PI 3.14 and #define MAX 100 create constants.
3. IDE = editor+compiler+debugger. Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 user screen, F8 step.
4. void invalid (keyword). std_marks valid. 1st number invalid (digit start + space).
5. TOTAL and total are different. Keywords: int, return, while.

Long questions (board Section C)

1. Explain the structure of a C program with a labelled example. Discuss tokens: keywords, identifiers, constants, operators.

Data types, variables, constants and input/output ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- List C data types and format specifiers
- Use printf, scanf, getchar, getch, puts, gets (exam style)
- Convert algebraic expressions into C

Type	Typical size (Turbo C 16-bit)	Specifier	Example
char	1 byte	%c	grade = 'A'
int	2 bytes	%d	n = 25
float	4 bytes	%f	avg = 72.5
double	8 bytes	%lf	pi = 3.14159
long	4 bytes	%ld	pop = 16000000L

Constants: 25 integer, 3.14 float, 'A' character, "BIEK" string. Variables must be declared before use: int roll; float marks;

```
int a, b, sum;
printf("Enter two numbers: ");
scanf("%d%d", &a, &b); /* & is the address operator */
sum = a + b;
printf("Sum = %d\n", sum);
```

Function	Use
printf	Formatted output to screen
scanf	Formatted input; needs & for scalars
getchar	Read one character (waits for Enter)
getch	Read one character without echo (conio.h)
gets / puts	String line I/O (unsafe in modern C, still in old papers)

Algebra → C (every year)

Mathematics	C expression
$A = \frac{1}{2} \times \text{base} \times \text{height}$	<code>A = 0.5 * base * height;</code>
$x = 3ab^3 + 3a^2b$	<code>x = 3*a*b*b*b + 3*a*a*b;</code>
$d = b^2 - 4ac $	<code>d = fabs(b*b - 4*a*c); /* math.h */</code>
$A = \pi r^2$	<code>A = 3.14 * r * r;</code>
$(a+b)^2 / ab$	<code>((a+b)*(a+b))/(a*b)</code>
$y = (a+b)^n$	<code>y = pow(a+b, n);</code>

WORKED EXAMPLE / BOARD STYLE

If $a=2$, $b=5$, $c=10$:

$200*((a*10)+c)$ wait — follow the paper's parentheses exactly.

$a*b*10/c+a+c = 2*5*10/10+2+10 = 100/10+2+10 = 10+2+10 = 22$ (integer division if all int).

$a-(c/b)/2+100 = 2-(10/5)/2+100 = 2-2/2+100 = 2-1+100 = 101$ with int division of $10/5$ first.

COMMON BOARD MISTAKE

`scanf("%d", &n)` — forgetting `&` is the #1 runtime bug. `%` with float is undefined; `%` is for integers only.

اردو میں سمجھیں

int عدد صحیح، float اعشاری، char ایک حرف، `%%`، `%f`، `%c`، `%d` فارمیٹ۔ `scanf` میں `&` پتہ دیتا ہے۔
ریاضی کو C میں `*` / اور `pow()` سے لکھیں۔ $\frac{1}{2}$ کو `*0.5` بنائیں۔

CHECK YOURSELF

1. Format specifier for int?

2. Why `&` in `scanf`?

Answers: 1. `%d` · 2. Pass the address so the value can be stored in the variable

MCQs

1. `%` operator is only used with:

- a. float
- b. int
- c. double
- d. string

2. The address operator is:

- a. `*`
- b. `&`
- c. `%`
- d. `#`

Key: 1=b, 2=b

Short questions (3 marks each)

1. How does `scanf()` differ from `getchar()`?
2. What is the address operator? Explain with example.
3. Convert $A = \frac{1}{2} \text{ base} \cdot \text{height}$ and $A = \pi r^2$ into C.
4. Define increment and decrement operators.

Model short answers

1. scanf reads formatted data (numbers, strings) using specifiers; getchar reads a single character from the keyboard buffer.
2. & gives the memory address. `int x=5; printf("%u", &x);` prints the address of x. `scanf("%d", &x)` stores input at that address.
3. `A=0.5*base*height;` `A=3.14*r*r;` (or `PI*r*r`).
4. ++ adds 1, -- subtracts 1. Pre (++k) changes then uses; post (k++) uses then changes.

Long questions (board Section C)

1. What are different types of operators in C? Explain with examples. Also discuss printf and scanf.

Operators, precedence and expressions ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Classify arithmetic, relational, logical, assignment, increment, conditional, bitwise operators
- Apply precedence: () then ++/-- then * / % then + -
- Trace small programs with x++ vs ++x

Class	Operators	Result
Arithmetic	+ - * / %	% remainder, integers only
Relational	> < >= <= == !=	1 (true) or 0 (false)
Logical	&& !	compound conditions
Assignment	= += -= *= /= %=	x += 2 means x = x+2
Increment	++ --	pre or post
Conditional	? :	x = (a>b)? a : b;
Bitwise	& ^ ~ << >>	bit level (rare in Paper II but listed in books)

LEARN IT

Precedence (high → low): () ++ -- * / % + - relational logical assignment. Same-level * / % go left to right.

Integer division: 5/2 is 2, not 2.5. Use 5/2.0 or a float variable.

WORKED EXAMPLE / BOARD STYLE

If x = 1:

printf("x=%d\n", x); → x=1

printf("x=%d\n", x + x); → x=2

printf("x=%d\n", x); → x=1 (x was never assigned a new value)

Post-increment example: k++ is post; ++k is pre. MCQ: k++ is post increment.

```
int a = 5, b = 2, c;
c = a / b;           /* 2 */
c = a % b;           /* 1 */
c = (a > b) && (b > 0); /* 1 */
c = (a > b) ? a : b;  /* 5 */
```

HOW BIEK ASKS THIS

Long 2026 Q4: types of operators in C. Make a table with one example each. Trace questions: never change a variable unless you see =, ++ or --.

CHECK YOURSELF

1. Value of 17 % 5?
2. Is k++ pre or post increment?

Answers: 1. 2 · 2. Post

MCQs

1. An example of post increment is:
 - a. k++
 - b. ++k
 - c. k+=
 - d. k=+
2. Highest precedence among + - / % is:
 - a. +
 - b. -
 - c. /
 - d. % — actually / and % are equal and higher than + -
3. == is a:
 - a. assignment
 - b. relational operator
 - c. loop
 - d. header

Key: 1=a, 2=c, 3=b

Short questions (3 marks each)

1. Define increment and decrement with examples.
2. If x=1, output of three printf lines using x, x+x, x.
3. Write C expressions for $(x^2 - y^2)(x+y)/(a+b)$.

Model short answers

1. ++k pre-increment, k++ post. -- same for decrement. int k=5; printf("%d", k++); prints 5 then k becomes 6.
2. 1 then 2 then 1.
3. $((x*x - y*y)*(x+y))/(a+b)$

Long questions (board Section C)

1. What are different types of operators used in a C program? Explain with examples and precedence.

Selection: if, if-else, nested if and switch ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write if, if-else, else-if ladder and nested if
- Write switch-case with break and default
- Draw the switch logical diagram
- Program: greater of two, pass/fail, leap year

```
if (marks ≥ 40)
    printf("Pass");
else
    printf("Fail");
```

```
if (n > 0) printf("Positive");
else if (n < 0) printf("Negative");
else printf("Zero");
```

switch (board long question)

```
int ch;
printf("1.Add 2.Sub 3.Mul\n");
scanf("%d", &ch);
switch (ch) {
    case 1: printf("Add"); break;
    case 2: printf("Sub"); break;
    case 3: printf("Mul"); break;
    default: printf("Invalid");
}
```

LEARN IT

switch works on int or char (not float). Each case should end with break, otherwise execution falls through. default is optional but expected in board answers.

Evaluate expression → Match case → Execute statements → break → leave switch

WORKED EXAMPLE / BOARD STYLE

Greater of two using a function (2026 Q2 xii):

```
int greater(int x, int y) {
    if (x > y) return x;
    else return y;
}
void main() {
    int a, b;
    scanf("%d%d", &a, &b);
    printf("Greater = %d", greater(a, b));
}
```

HOW BIEK ASKS THIS

2026 Section C Q3: Define switch with logical diagram. Draw a diamond or a multi-way box: expression, case 1, case 2, default.

اردو میں سمجھیں

if شرط سچی ہو تو بلاک چلتا ہے۔ switch ایک متغیر کی کئی قیمتوں پر چلتا ہے، ہر case کے بعد break لکھیں۔

CHECK YOURSELF

1. Can switch test a float?
2. What if you omit break?

Answers: 1. No · 2. Fall-through: the next case also runs

MCQs

1. switch is a:
 - a. loop
 - b. selection / decision structure
 - c. function
 - d. header
2. default in switch runs when:
 - a. always
 - b. no case matches
 - c. every case
 - d. at compile time

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define switch with a small example.
2. Write a program that prints the greater of two integers using a function.
3. Differentiate if-else and switch.

Model short answers

1. switch(expression) { case c1: ... break; default: ... } selects one case by matching integers/chars.
2. Full greater() program as in the lecture.
3. if-else handles ranges and floats; switch is multi-way on discrete int/char values and is cleaner for menus.

Long questions (board Section C)

1. Define switch statement with a logical diagram and a complete C program (menu or grade).

Loops: for, while, do-while, break, continue, patterns ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write for, while and do-while and compare them
- Use break and continue
- Print multiplication tables, factorial, nested-loop patterns

	for	while	do-while
Test	At the top	At the top	At the bottom — runs at least once
Best when	Count known	Condition, count unknown	Menus, "repeat until"
Header	for(s;c;u)	while(c)	do { } while(c);

```
/* table of n */
int n, i;
scanf("%d", &n);
for (i = 1; i ≤ 10; i++)
    printf("%d x %d = %d\n", n, i, n * i);
```

```
/* factorial : 4! = 24 */
int n, i; long f = 1;
scanf("%d", &n);
for (i = 1; i ≤ n; i++)
    f = f * i;
printf("%ld", f);
```

break vs continue

- break: leave the loop immediately.
- continue: skip the rest of this iteration and go to the next test / increment.

```
int i;
for (i = 1; i ≤ 5; i++) {
    if (i == 3) continue;
    printf("%d ", i);    /* 1 2 4 5 */
}
```

Nested loops — 2026 pattern

```
int i, j;
for (i = 1; i ≤ 5; i++) {
    for (j = 1; j ≤ i; j++)
        printf("%d", j);
    printf("\n");
}
/* 1
12
123
1234
12345 */
```

Number / Square: `for(i=1;i<=5;i++) printf("%d %d\n", i, i*i);`

HOW BIEK ASKS THIS

Programs to memorise: table, factorial, even 0–50, sum of n numbers, reverse digits, prime check, Fibonacci, patterns.

COMMON BOARD MISTAKE

do-while ends with a semicolon after `while(condition);` forgetting it is a syntax error. Infinite loop: `for(;;)` or `while(1)`.

اردو میں سمجھیں

for گنتی معلوم ہو تو، while شرط پر، do-while کم از کم ایک بار۔ break لوپ توڑتا ہے، continue اس چکر کو چھوڑتا ہے۔
ٹیبل، فیکٹوریل اور پیٹرن ہر سال آتے ہیں — زبانی لکھ کر یاد کریں۔

CHECK YOURSELF

1. $4! = ?$
2. Does do-while always run once?

Answers: 1. 24 · 2. Yes

MCQs

1. `for()` is used when we need:
 - a. Sequential structure
 - b. Selection structure
 - c. Iteration structure
 - d. Top down structure
2. This statement skips the rest of the loop body and continues from the top:
 - a. `break`
 - b. `resume`
 - c. `continue`
 - d. `skip`

Key: 1=c, 2=c

Short questions (3 marks each)

1. Write a program that prints the multiplication table of an inputted number.
2. Write a program that prints factorial of n (hint $4!=24$).
3. Differentiate while and do-while.
4. Define break and continue with an example.
5. Write a nested-loop program for the triangle 1 / 12 / 123 / ...

Model short answers

1. for i=1 to 10 print n x i = n*i.
2. f=1; for i=1..n f*=i; print f.
3. while tests first (may run 0 times). do-while tests after the body (at least once).
4. break exits; continue jumps to next iteration. Show a 4-line for-loop example.
5. Outer i=1..5, inner j=1..i print j, then newline.

Long questions (board Section C)

1. What is a loop? Explain for, while and do-while with programs. Discuss nested loops.

Arrays and strings ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Declare, initialise and traverse 1-D and 2-D arrays
- Explain string terminator '\0' and strlen, strcpy, strcat, strcmp
- Predict output of num[3] style MCQs

Array. A collection of elements of the same data type stored in contiguous memory, accessed by index starting at 0.

```
int num[] = {3, 5, 7, 10, 20};
printf("%d", num[3]);    /* 10 - index 3 is the fourth element */
```

```
int a[5], i, sum = 0;
for (i = 0; i < 5; i++) {
    scanf("%d", &a[i]);
    sum += a[i];
}
printf("Average = %.2f", sum / 5.0);
```

Two-dimensional array

```
int m[3][3], i, j;
for (i = 0; i < 3; i++)
    for (j = 0; j < 3; j++)
        scanf("%d", &m[i][j]);
```

Strings

A string is a char array ending with '\0'. char name[20] = "Ali"; /* 'A','l','i','\0' */

Function	Header	Use
strlen(s)	string.h	length without '\0'
strcpy(d,s)	string.h	copy s into d
strcat(d,s)	string.h	join s onto d
strcmp(a,b)	string.h	0 if equal, <0 or >0 if not

```
char s[30];
int i, n;
gets(s);          /* exam style */
n = strlen(s);
for (i = n - 1; i ≥ 0; i--)
    printf("%c", s[i]); /* reverse name */
```

HOW BIEK ASKS THIS

2025: What is an array? Declare and initialise. Output of num[3] with {3,5,7,10,20} is 10 — not 7. Students forget index 0.

COMMON BOARD MISTAKE

scanf("%s", name) — no & for arrays because the name is already an address. Overrunning the array size crashes the program.

اردو میں سمجھیں

Array ایک ہی قسم کی کئی قیمتیں، انڈیکس 0 سے۔ `num[3]` چوتھا عنصر ہے۔
سٹرنگ حروف کی صف ہے جو `\0` پر ختم ہوتی ہے۔ `strlen` لمبائی، `strcpy` نقل، `strcat` جوڑ، `strcmp` موازنہ۔

CHECK YOURSELF

1. Index of the first element?

2. `num[]={3,5,7,10,20}; num[3]?`

Answers: 1. 0 · 2. 10

MCQs

1. Array index in C starts from:

- a. 1
- b. 0
- c. -1
- d. 2

2. A string in C ends with:

- a. `\n`
- b. `\0`
- c. `\t`
- d. NULL pointer only

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is an array? Declare and initialise an integer array.
2. `int num[]={3,5,7,10,20};` what does `printf("%d", num[3])` print?
3. Write a program to reverse an inputted name using `strlen()`.
4. Name four string functions.

Model short answers

1. Same-type contiguous list. `int a[5]={1,2,3,4,5};` or `int a[]={10,20};`
2. 10.
3. Read string, loop from `strlen-1` down to 0 printing characters.
4. `strlen`, `strcpy`, `strcat`, `strcmp` (plus `strncpy` etc. if space).

Long questions (board Section C)

1. Explain array and its types with examples. Write a program to add two 3×3 matrices.

Functions: prototype, call, return, types ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- State advantages and characteristics of functions
- Differentiate built-in and user-defined; call by value vs reference
- Write programs that use functions (greater, factorial, area)

Function. A named, reusable block that performs a task. `main()` is itself a function. Advantages: modularity, reuse, easier testing, team work.

Prototype (declaration) → Definition (body) → Call (use)

```
int square(int n);           /* prototype */
void main() {
    printf("%d", square(5)); /* call – actual parameter 5 */
}
int square(int n) {          /* definition – n is formal parameter */
    return n * n;
}
```

Idea	Meaning
Built-in	printf, scanf, sqrt, strlen — already in libraries
User-defined	You write them
Return type	void if nothing is returned
Formal parameters	In the definition header
Actual parameters	In the call
Call by value	A copy is passed; original safe
Call by reference	Address (&) / pointer; function can change the original
Local vs global	Local inside block; global outside functions
Recursion	Function calls itself (factorial, Fibonacci)

HOW BIEK ASKS THIS

2026 long Q5: main characteristics of a function with example. List: name, return type, parameters, independent compilation, reuse, call by value/reference.

اردو میں سمجھیں

فنکشن نام والا بلاک ہے جو بار بار کام کرتا ہے۔ Prototype اعلان، تعریف جسم، کال استعمال۔ return قدر بھیجتا ہے۔

CHECK YOURSELF

1. Does main have to exist?
2. sqrt comes from which header?

Answers: 1. Yes — it is the entry point · 2. math.h

MCQs

1. A subprogram that returns a value is a:
 - a. function
 - b. report
 - c. macro only
 - d. file
2. Passing a copy of a variable is:
 - a. pass by value
 - b. pass by address
 - c. pass by reference
 - d. pass by pointer

Key: 1=a, 2=a

Short questions (3 marks each)

1. Write any one program using a function to print the greater of two numbers.
2. Differentiate formal and actual parameters.
3. What are built-in functions? Give three.

Model short answers

1. `int greater(int x,int y){ return (x>y)?x:y; }` call from main after `scanf`.
2. Formal: names in the definition. Actual: values in the call. They correspond in order and type.
3. Library functions: `printf`, `scanf`, `getchar`, `strlen`, `sqrt`.

Long questions (board Section C)

1. What are the main characteristics of a function? Describe with an example program.

Pointers and the address operator ★ Golden

One college period · 40–45 minutes · Pointers (Sindh XII + C paper)

What you will learn

- Explain & and *
- Declare and initialise a pointer
- Write a program that prints address and value
- Pass addresses to a function to double two numbers

Pointer. A variable that stores the memory address of another variable. Declared with *. The address operator & gives an address; the dereference operator * accesses the value at that address.

```
int x = 10;
int *p;      /* p is a pointer to int */
p = &x;      /* p holds the address of x */
printf("Value = %d\n", *p);  /* 10 */
printf("Address = %u\n", p);
```

```
void double_them(int *a, int *b) {
    *a = *a * 2;
    *b = *b * 2;
}
void main() {
    int x = 4, y = 5;
    double_them(&x, &y);
    printf("%d %d", x, y);  /* 8 10 */
}
```

HOW BIEK ASKS THIS

2025 Paper II: What is a pointer in C? How is it initialised? What is the address operator?

COMMON BOARD MISTAKE

p = &x is wrong for the first assignment if p is int. Use p = &x; then read *p. Mixing them loses marks.

اردو میں سمجھیں

پوائنٹر وہ متغیر ہے جو دوسرے متغیر کا پتہ رکھتا ہے۔ & پتہ دیتا ہے، * اس پتے پر موجود قدر دیتا ہے۔

CHECK YOURSELF

1. If `int x=5; int *p=&x;` what is `*p`?
2. Symbol for address operator?

Answers: 1. 5 · 2. &

MCQs

1. A pointer stores a:
 - a. value only

- b. memory address
- c. file name
- d. keyword

2. &x means:

- a. value of x
- b. address of x
- c. pointer declaration
- d. multiply x

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is a pointer in C? How is it initialised?
2. Explain the address operator with an example.
3. Write a program to display address and value of a variable using a pointer.

Model short answers

1. A variable that holds an address. `int *p; p = &x;` or `int *p = &x;`
2. `&x` is the address of x. `scanf` uses it; pointers store it.
3. Declare x and `p=&x;` `printf` value `*p` and address `p`.

Long questions (board Section C)

1. Explain pointers with a diagram of x and p. Write a function that receives two addresses and doubles both values.

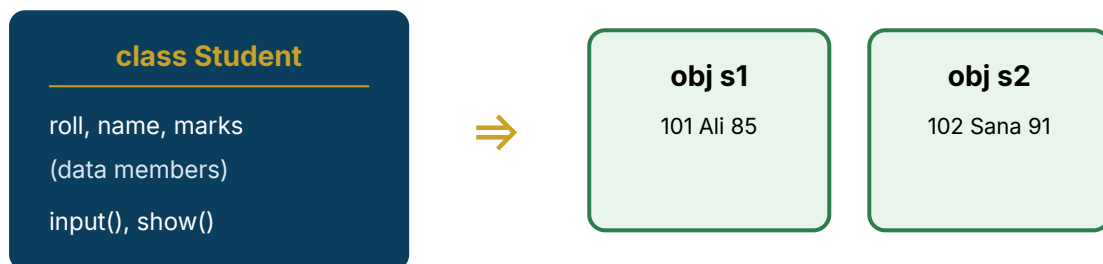
OOP in C++: class, object, constructor, inheritance, polymorphism

★ Golden

One college period · 40-45 minutes · OOP using C++ and file handling

What you will learn

- State encapsulation, inheritance, polymorphism
- Write a simple class Student with constructor
- Explain public, private, protected
- Open text/binary files for read, write, append



A class is a blueprint; objects are instances with their own data.

- Encapsulation: wrap data + functions; hide data with private.
- Inheritance: derived class reuses base class (reusability).
- Polymorphism: one name, many forms — overload (same name, different parameters) and override (same signature in derived class).

```
#include <iostream>
using namespace std;
class Student {
    int age; float percentage;           // private by default
public:
    Student() { age = 0; percentage = 0; }           // default constructor
    Student(int a, float p) { age = a; percentage = p; } // overloaded
    void show() { cout << age << " " << percentage; }
    ~Student() { } // destructor
};
int main() {
    Student s1(17, 82.5);
    s1.show();
}
```

Access specifiers: private (only inside class), public (everywhere), protected (class + derived).

Constructor has the class name, no return type, called automatically. Destructor ~ClassName() at end of life.

Inheritance sketch

```
class Base { public: int x; };
class Drive : public Base { public: int y; };
// Drive object can use x and y
```

File handling

Text files store characters; binary files store raw bytes (objects). Modes: ios::in read, ios::out write (truncate), ios::app append. BOF beginning, EOF end. Streams: get/put (char), getline (string), read/write (binary).

```
#include <fstream>
ofstream out("data.txt");
out << "Ali 80\n";
out.close();
ifstream in("data.txt");
string line; getline(in, line);
```

HOW BIEK ASKS THIS

Sindh lab: class STUDENT with AGE and PERCENTAGE; TIME class with constructors; inherit DRIVE from BASE; write/read text file; write/read student object as binary.

اردو میں سمجھیں

کلاس خاکہ ہے، آبجیکٹ اس کی کاپی۔ Encapsulation ڈیٹا چھپاتا ہے، Inheritance وراثت، Polymorphism ایک نام کئی کام۔
Constructor آبجیکٹ بنتے وقت چلتا ہے۔
فائل بینڈلنگ: ٹیکسٹ اور بائنری، read/write/append.

CHECK YOURSELF

1. Does a constructor return a value?
2. ios::app means?

Answers: 1. No — not even void · 2. Append to the file without erasing old data

MCQs

1. Wrapping data and functions together is:
 - a. inheritance
 - b. encapsulation
 - c. compilation
 - d. paging
2. A constructor has:
 - a. the class name and no return type
 - b. any name
 - c. return type int
 - d. a destructor symbol

Key: 1=b, 2=a

Short questions (3 marks each)

1. Explain class and object.
2. Differentiate constructor and destructor.
3. Differentiate text and binary files.

Model short answers

1. Class: user-defined type with data members and member functions. Object: variable of that type.
2. Constructor initialises (same name, auto-called). Destructor cleans up (~Name), no arguments.
3. Text: human-readable characters. Binary: exact memory image, efficient for objects.

Long questions (board Section C)

1. Explain OOP principles with a C++ class that uses a constructor. Discuss inheritance and polymorphism briefly.

Visual Basic IDE, forms and ActiveX controls ★ Golden

One college period · 40–45 minutes · Programming using Visual Basic

What you will learn

- Describe the VB IDE and a form
- List basic ActiveX controls and important properties
- Set Cancel button, use labels, text boxes, command buttons, combo, option, check

Visual Basic. An event-driven programming language and IDE from Microsoft. You draw controls on a Form; code runs when events occur (Click, Load, Change).

Control	Use	Key properties
Form	Window	Caption, BackColor, BorderStyle
Label	Display text the user does not type	Caption, Font, AutoSize
TextBox	Input / output of text	Text, MaxLength, PasswordChar, MultiLine
CommandButton	Click to run code	Caption, Enabled, Cancel, Default
CheckBox	Yes/No independent	Value 0/1/2
OptionButton	One choice from a group	Value True/False
ComboBox / ListBox	Pick from a list	List, Text, AddItem
Image / PictureBox	Pictures	Picture, Stretch

Cancel button property belongs to CommandButton (MCQ). Default step in For...Next is 1. DIM declares a variable. Variant can hold any type: Dim x As Variant or Dim x (undeclared = Variant if no Option Explicit — board still asks “how can a variable be defined as variant”).

```
Private Sub cmdAdd_Click()
    lblSum.Caption = Val(txtA.Text) + Val(txtB.Text)
End Sub
```

HOW BIEK ASKS THIS

2026 shorts: ComboBox properties; TextBox properties; basic ActiveX controls; Cancel button object; DIM; Option Button vs CheckBox.

اردو میں سمجھیں

Visual Basic میں فارم پر بٹن، ٹیکسٹ باکس، لیبل رکھتے ہیں۔ کلک ایونٹ پر کوڈ چلتا ہے۔ Option Button ایک انتخاب، CheckBox ہاں/نہیں۔

CHECK YOURSELF

1. Default For step?

2. Which control is for a single choice in a group?

Answers: 1. 1 · 2. Option Button (radio)

MCQs

1. The Cancel button property belongs to:
 - a. Form
 - b. Label
 - c. Button
 - d. Textbox
2. Default Step in For...Next is:
 - a. 2
 - b. 0
 - c. -1
 - d. 1
3. DIM declares a:
 - a. variable
 - b. constant only
 - c. keyword
 - d. function only
4. A control that selects a single choice from a group:
 - a. Check Box
 - b. Option Button
 - c. Caption Box
 - d. List Box

Key: 1=c, 2=d, 3=a, 4=b

Short questions (3 marks each)

1. What is a ComboBox? List properties.
2. Write properties of a TextBox.
3. What are basic ActiveX controls in VB?
4. How can a variable be defined as Variant?

Model short answers

1. Drop-down list combining text box + list. Properties: List, ListIndex, Text, Style, AddItem, Sorted.
2. Text, Font, MaxLength, PasswordChar, MultiLine, Locked, Alignment.
3. Label, TextBox, CommandButton, CheckBox, OptionButton, ComboBox, ListBox, Frame, Timer, Image.
4. Dim x As Variant or Dim x (Variant is the default type).

Long questions (board Section C)

1. Explain commonly used VB controls with properties. Write a form program that adds two numbers.

VB language: operators, if, Select Case, loops, arrays, functions

★ Golden

One college period · 40–45 minutes · Programming using Visual Basic

What you will learn

- Write If...Then and Select Case
- Write For...Next, Do While, Do Until
- Declare arrays and small functions
- Translate formulae into VB

```
If marks ≥ 40 Then
    lblR.Caption = "Pass"
Else
    lblR.Caption = "Fail"
End If
```

```
Select Case ch
    Case 1: lbl.Caption = "Add"
    Case 2: lbl.Caption = "Sub"
    Case Else: lbl.Caption = "Invalid"
End Select
```

```
Dim i As Integer, s As Integer
s = 0
For i = 0 To 50 Step 2
    s = s + i          ' even numbers 0-50
Next i
```

```
' sum and average of 10 inputted numbers
Dim i As Integer, n As Double, t As Double
t = 0
For i = 1 To 10
    n = Val(InputBox("Number " & i))
    t = t + n
Next i
MsgBox "Sum=" & t & " Avg=" & t / 10
```

Maths	VB
$x = 3ab^2 / c$	<code>x = 3 * a * b ^ 2 / c</code>
$Y = (4/3)\pi r^3$	<code>Y = (4 / 3) * 3.14 * r ^ 3</code>
$Y = (a+b)^n$	<code>Y = (a + b) ^ n</code>
$a = \frac{1}{2} \text{ base} \cdot \text{height}$	<code>a = 0.5 * base * height</code>

Built-in functions: Val, Str, Left, Right, Mid, Len, UCase, LCase, Int, Rnd, Date, Sqr, Abs, Format. User-defined: Function Area(r As Double) As Double : Area = 3.14 * r * r : End Function. Errors: syntax, run-time, logical. On Error GoTo handler.

HOW BIEK ASKS THIS

2026 programs: even 0–50; sum and average of ten numbers. Long: arrays; loops; built-in function with a program.

اردو میں سمجھیں

VB میں If...Then، Select Case، For...Next اور Do While لوپ ہیں۔ ^ طاقت ہے۔ Val ٹیکسٹ کو عدد بناتا ہے۔

CHECK YOURSELF

1. Even numbers 0–50: Step?

2. Val("12a") in VB6?

Answers: 1. 2, start 0 · 2. 12 (stops at first non-numeric, classic VB)

MCQs

- Comparison operators are also called:
 - Comparators
 - Comparables
 - Relations
 - Relational
- The last statement in a Function is often:
 - Footer
 - Header
 - End Function / Return value assignment
 - Call
- A function returns a:
 - form
 - value
 - only a MsgBox
 - table

Key: 1=d, 2=c, 3=b

Short questions (3 marks each)

- Write the structure of Select Case.
- Write a VB program to generate even numbers 0–50.
- Write a VB program to calculate sum and average of ten numbers.
- Convert $Y = (4/3)\pi r^3$ to VB.
- What is an error? Define its types.

Model short answers

- Select Case expr : Case v1: ... : Case Else: ... : End Select
- For i = 0 To 50 Step 2 : Print i : Next
- Loop 10 InputBox, add to total, divide by 10.
- $Y = (4 / 3) * 3.14 * r ^ 3$
- Mistake in a program. Syntax (grammar), run-time (divide by zero), logical (wrong formula, runs but answer wrong).

Long questions (board Section C)

1. Explain arrays in VB with types and an example. OR Explain loops with two types and programs. OR Built-in functions with a program.

DBMS, models, ER diagrams, keys and normalisation ★ Golden

One college period · 40–45 minutes · Database fundamentals

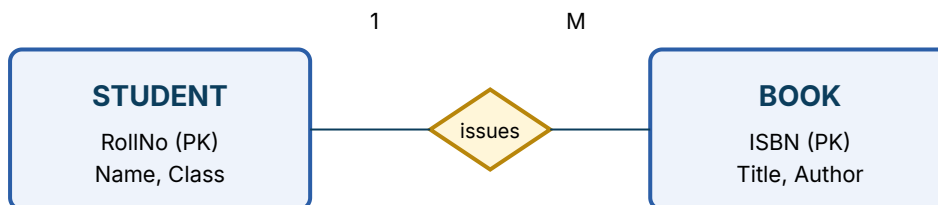
What you will learn

- Differentiate file system and DBMS
- Draw hierarchical, network, relational, object models
- Define PK, FK, candidate, super key; 1NF 2NF 3NF
- Draw a simple ER diagram (library / student-book)

Database. An organised collection of related data. DBMS is software that creates, manages and controls access to the database (MS Access, Oracle, MySQL).

Advantages vs flat files: less redundancy, data integrity, sharing, security, backup, query language. Components: hardware, software, data, procedures, people, DML/DDL (data access language). Users: DBA, application programmer, end user.

Model	Picture
Hierarchical	Tree — one parent, many children (IMS)
Network	Graph — many-to-many with pointers (CODASYL)
Relational	Tables (relations) linked by keys — SQL works here
Object / object-relational	Objects with methods; ORDBMS



Entity Relationship: rectangles = entities, diamond = relationship, ovals = attributes (in full ER).

Term	Meaning
Table / relation	A grid of rows and columns
Record / tuple	A row
Field / attribute	A column
Primary key	Unique identifier, not null
Foreign key	A field that matches the PK of another table
Candidate key	Any field(s) that could be PK
Super key	Any set that uniquely identifies, may have extras
1NF	Atomic cells, no repeating groups
2NF	1NF + non-key fields depend on whole PK
3NF	2NF + no transitive dependence on non-keys

Relationships: 1:1, 1:M, M:M, M:1. Integrity: NOT NULL, PRIMARY KEY, FOREIGN KEY (referential). Three ANSI levels: internal/physical, conceptual/logical, external/view.

HOW BIEK ASKS THIS

Both options 2026: PK vs FK; table, query, report; DBMS components; data models long question; Access data types long question.

اردو میں سمجھیں

ڈیٹا بیس منظم متعلقہ ڈیٹا ہے۔ DBMS سافٹ ویئر (Access) اسے سنبھالتا ہے۔ Primary Key یکتا، Foreign Key دوسری ٹیبل کا لنک۔

Relational ماڈل ٹیبلز اور SQL۔ 1NF 2NF 3NF ڈیٹا دہرانے سے بچاتے ہیں۔

CHECK YOURSELF

1. Row is also called?
2. SQL works on which model?

Answers: 1. Record / tuple · 2. Relational

MCQs

1. A unique key in a table is the:
 - a. Primary key
 - b. Foreign key
 - c. Super key only
 - d. Candidate only — PK is the chosen candidate
2. The primary object in a database is the:
 - a. Table
 - b. Query
 - c. Report
 - d. Form
3. Tree-like database model:

- a. Relational
- b. Hierarchical
- c. Network
- d. Object

4. Columns in a database are:

- a. Tuples
- b. Attributes
- c. Keys only
- d. Tables

5. The model that allows SQL is:

- a. Network
- b. Hierarchical
- c. Relational
- d. Flat

6. DBMS advantage:

- a. data depends on programs
- b. redundancy increases
- c. data is integrated and shared
- d. integrity decreases

Key: 1=a, 2=a, 3=b, 4=b, 5=c, 6=c

Short questions (3 marks each)

1. Differentiate primary key and foreign key.
2. Define table, query, report.
3. Name the major components of DBMS.
4. Who is a DBA? Three responsibilities.
5. Define one-to-many relationship.
6. Write any four advantages of DBMS.
7. Name four objects in MS Access.

Model short answers

1. PK uniquely identifies a row in its own table. FK is a field pointing to a PK in another table to link records.
2. Table stores data. Query asks questions / filters. Report is a printable layout of data.
3. Hardware, software, data, procedures, users, query language.
4. Database Administrator: security, backup, performance, user accounts, recovery.
5. One parent row links to many child rows (one class, many students). Draw 1 — M.
6. Less redundancy, integrity, sharing, security, standard SQL, backup.
7. Tables, Queries, Forms, Reports (Macros, Modules).

Long questions (board Section C)

1. What is a data model? Explain types with diagrams. OR Explain ER modelling and normalisation up to 3NF with a library example.

MS Access objects, data types and SQL ★ Golden

One college period · 40–45 minutes · Database fundamentals

What you will learn

- List Access objects and data types for a long answer
- Write CREATE, INSERT, UPDATE, DELETE, SELECT
- Plan a Library Management schema

Object	Purpose
Table	Store data in fields/records; set PK
Query	Select, update, calculate, join tables
Form	On-screen data entry (like a VB form)
Report	Print marksheets, lists
Macro / Module	Automate / VBA

MS Access data types (learn for Section C)

Type	Stores
Text / Short Text	Names, up to 255 chars
Memo / Long Text	Remarks, long notes
Number	Byte, Integer, Long, Single, Double
Date/Time	Dates and clocks
Currency	Money, 4 decimal
AutoNumber	Automatic PK
Yes/No	Boolean
OLE Object	Pictures, files
Hyperlink	URLs
Attachment / Lookup	Newer Access extras

```

CREATE TABLE Book (
  ISBN TEXT PRIMARY KEY,
  Title TEXT,
  Author TEXT,
  Price CURRENCY
);
INSERT INTO Book VALUES ('978-1', 'CS XI', 'STBB', 450);
SELECT Title, Author FROM Book WHERE Price > 400;
UPDATE Book SET Price = 500 WHERE ISBN = '978-1';
DELETE FROM Book WHERE ISBN = '978-1';

```

SELECT is projection (columns) + WHERE is selection (rows). JOIN combines tables on keys. Equijoin: WHERE Book.AuthorID = Author.ID. Operators: AND OR NOT, = <> > < LIKE BETWEEN.

Planning a library database

1. Purpose: issue and return books.
2. Tables: STUDENT(Roll PK, Name, Class), BOOK(ISBN PK, Title, Author), ISSUE(IssueID PK, Roll FK, ISBN FK, IssueDate, ReturnDate).
3. Unique fields: Roll, ISBN, IssueID.
4. Relationships: Student 1—M Issue M—1 Book, all up to 3NF (no repeating books inside student).
5. Refine: add Fine, status Issued/Returned.

HOW BIEK ASKS THIS

Practical 2026 Access: create Library_Management, PK, 5 records, query currently issued books.
Theory: explain all Access data types.

اردو میں سمجھیں

Access میں Table ڈیٹا، Query سوال، Form انٹری، Report پرنٹ، SQL: SELECT چننا، INSERT داخل، UPDATE بدل، DELETE مٹانا۔

CHECK YOURSELF

1. Which object is for printing?
2. AutoNumber is typically used as?

Answers: 1. Report · 2. Primary key

MCQs

1. MS Access object used to display and print data:
 - a. Report
 - b. Table
 - c. Form
 - d. Macro
2. This key uniquely identifies a record:
 - a. Super Key
 - b. Foreign Key
 - c. Conditional Key
 - d. Primary Key

Key: 1=a, 2=d

Short questions (3 marks each)

1. Write SELECT syntax with WHERE.
2. Differentiate form and report.
3. Write steps to design a library database.

Model short answers

1. SELECT field1, field2 FROM table WHERE condition; example SELECT Title FROM Book WHERE Price>400;
2. Form is interactive on screen; report is formatted for paper/PDF.
3. Purpose, tables, fields, unique keys, relationships to 3NF, refine.

Long questions (board Section C)

1. Explain all the data types used in MS-Access. OR Explain SQL DML statements with examples.

SDLC, stakeholders and phases ★ Golden

One college period · 40–45 minutes · System Development Life Cycle

What you will learn

- Define system and SDLC
- Name stakeholder roles
- Describe planning, analysis, design, coding, testing, deployment, maintenance

System. A set of related components that work together toward a goal. SDLC is the planned process of building information software from idea to retirement.

Stakeholder	Role
Project manager	Plan, cost, schedule
Business analyst	Gather requirements
System architect	Overall design
Developer	Write code
Tester	Find bugs
Operations	Deploy on servers
Production support	Maintenance after go-live

Planning → Requirement analysis → Design → Coding → Testing → Deployment →

Maintenance

- Planning: problem, feasibility (technical, economic, operational), schedule.
- Requirement engineering: functional (what it must do) vs non-functional (speed, security). SRS document.
- Design: architecture, algorithms, flowcharts, database schema, UI.
- Implementation: coding in C / VB / other.
- Testing: find bugs and errors; unit, integration, system, acceptance.
- Deployment & SLA: install, train, service-level agreement.
- Maintenance: fix, adapt, improve.

Models you may mention: Waterfall (linear, old board favourite) vs Agile (iterative sprints).

اردو میں سمجھیں

SDLC سافٹ ویئر بنانے کا منصوبہ بند چکر ہے: منصوبہ، ضرورت، ڈیزائن، کوڈ، ٹیسٹ، تنصیب، دیکھ بھال۔

CHECK YOURSELF

1. SRS means?
2. Who gathers requirements?

Answers: 1. Software Requirements Specification · 2. Business analyst

MCQs

1. SDLC is mainly about:
 - a. repairing printers
 - b. developing software in stages
 - c. only testing viruses
 - d. drawing cables
2. Converting design into programs is:
 - a. testing
 - b. coding / implementation
 - c. feasibility
 - d. retirement

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define SDLC and list its phases.
2. Differentiate functional and non-functional requirements.
3. Who are the stakeholders of SDLC?

Model short answers

1. Life cycle of software: plan, analyse, design, code, test, deploy, maintain.
2. Functional: features (print marksheet). Non-functional: quality (response in 2 seconds, password security).
3. PM, analyst, architect, developers, testers, operations, support, plus the client/users.

Long questions (board Section C)

1. Explain the System Development Life Cycle with a diagram and the role of each stakeholder.

Multimedia systems; wireless and mobile communication

One college period · 40-45 minutes · Multimedia and wireless

What you will learn

- Define multimedia components and file types
- Use layers, keyframes, export/share a short video (theory + lab idea)
- Compare Wi-Fi, Bluetooth, infrared, WiMAX, microwave
- Outline cellular network, 2G-5G, GEO/MEO/LEO, ad hoc / MANET

Multimedia. The combination of text, graphics, animation, audio and video used to communicate. Advantages: richer teaching, advertising, games, training.

Component	Typical files
Text	.txt .doc .pdf
Graphics	.jpg .png .gif .bmp .svg
Animation	.gif .mp4
Audio	.mp3 .wav .aac
Video	.mp4 .avi .mkv

Packages: web apps vs desktop (ProShow Gold, JetAudio, Shotcut, Canva, PowerPoint). Scene/canvas = work area; layers stack pictures/text; keyframes mark motion in animation. Lab: 2-minute video, export MP4, upload/share.

Wireless

- Wireless: no physical cable; uses electromagnetic waves. Mobile: communication while moving.
- Terms: frequency spectrum, radio signal, transceiver, access point, line of sight.

Short range	Typical use
Wi-Fi	LAN internet in college / home
Bluetooth	earbuds, file share, few metres
Infrared	TV remote, line of sight
Microwave	dishes, backhaul
WiMAX	city-wide wireless (MAN-like)

Satellites: GEO ~36000 km (TV), MEO (GPS), LEO (phones, Starlink-style). Cellular: cell, BTS, BSC, MSC, frequency reuse. Generations: 2G GSM voice, 3G GPRS/data, 4G LTE fast data, 5G low latency. Ad hoc: no fixed AP; MANET is a mobile ad hoc network.

اردو میں سمجھیں

ملٹی میڈیا متن، تصویر، آواز، ویڈیو، اینیمیشن کا ملاپ ہے۔ وائرلیس تار کے بغیر ریڈیو لہریں: وائی فائی، بلوٹوتھ، سیلولر 2G سے 5G۔

CHECK YOURSELF

1. Bluetooth vs Wi-Fi range?
2. 4G technology name often asked?

Answers: 1. Bluetooth shorter; Wi-Fi room/building · 2. LTE

MCQs

1. MP3 is mainly a file of:
 - a. video
 - b. audio
 - c. database
 - d. source code
2. Wi-Fi is a:
 - a. wired LAN cable
 - b. short-range wireless technology
 - c. printer type
 - d. CPU register
3. GSM is associated with:
 - a. 2G
 - b. 5G only
 - c. fibre optics
 - d. OSI layer 7 only

Key: 1=b, 2=b, 3=a

Short questions (3 marks each)

1. Define multimedia and its components.
2. Differentiate Wi-Fi and Bluetooth.
3. What is a cellular network? Name 2G to 5G.
4. GEO vs LEO satellites.

Model short answers

1. Combination of text, graphics, animation, audio, video. Used in education and ads.
2. Wi-Fi: higher speed/range for internet. Bluetooth: very short range for accessories.
3. Geographic area split into cells with BTS. 2G GSM, 3G, 4G LTE, 5G.
4. GEO high and stationary over earth (TV). LEO low and fast (phones, imaging).

Long questions (board Section C)

1. Explain multimedia components and how you would make a 2-minute educational video. OR Explain wireless technologies and cellular generations.

Paper II Option I mock (C + Access) with answers ★ Golden

One college period · 40–45 minutes · Revision

What you will learn

- Sit a full Option I paper and check against keys

Section A key (15)

1 main() 2 C (Ritchie) 3 Programmer 4 Compiler 5 5number invalid 6 % with int 7 k++ post 8 / and %
highest among listed arithmetic 9 Primary key 10 continue 11 data integrated 12 Table 13 identifiers 14
Iteration 15 /* */

Section B — practise these 10

1. Skeleton of a C program.
2. #include and #define examples.
3. Program: table of n.
4. break vs continue.
5. Three reserved words.
6. Address operator example.
7. Factorial program.
8. scanf vs getchar.
9. PK vs FK.
0. Evaluate integer expressions.
1. DBMS components.
2. greater() function program.
3. Three maths → C conversions.
4. Nested pattern 1 / 12 / 123... and number-square table.
5. Table, query, report.

Section C — pick 3

1. switch with diagram.
2. Operators in C.
3. Characteristics of functions + program.
4. Data models in DBMS.
5. MS Access data types.

WORKED EXAMPLE / BOARD STYLE

Scoring a C program on the board: correct headers (1), main (1), declarations (1), logic/loop (3), output (1), indentation/comments (1). Always write clrscr/getch if your college uses Turbo C, and box the output.

اردو میں سمجھیں

آپشن I: C کے پروگرام کاغذ پر صحیح semicolon اور & کے ساتھ لکھیں۔ Access کے ڈیٹا ٹائپ زبانی یاد کریں۔

MCQs

1. Option I mock: the required function in every C program is:
 - a. main()
 - b. start()
 - c. loop()
 - d. file()

Key: 1=a

Long questions (board Section C)

1. Write the factorial program and the switch-menu program from memory in 25 minutes.

Paper II Option II mock (VB + Access) with answers ★ Golden

One college period · 40–45 minutes · Revision

What you will learn

- Sit a full Option II paper and check against keys

Section A key (15)

1 Command Button (Cancel) 2 Step default 1 3 Record 4 Report 5 Primary Key 6 Check Box is 0/1 (paper wording) / Option for exclusive — learn both 7 Relational operators 8 Pass by value 9 Return/End Function 10 Attributes 11 Hierarchical 12 Option Button 13 Function 14 Variable 15 Relational model

Section B list from 2026

1. Four advantages of DBMS.
2. ComboBox + properties.
3. Variant type.
4. One-to-many + diagram.
5. Four Access objects.
6. Select Case structure.
7. DBA responsibilities.
8. Basic ActiveX controls.
9. ADO MDI SQL DBA OLE full forms (any three).
0. TextBox properties.
1. Three VB formulae.
2. Types of errors.
3. Even numbers 0–50 program.
4. Sum and average of ten numbers.
5. PK vs FK.

Section C

1. Arrays and types with examples.
2. Loop — any two types with examples.
3. Built-in function + program.
4. All database models with diagrams.
5. All MS Access data types.

WORKED EXAMPLE / BOARD STYLE

ADO = ActiveX Data Objects. MDI = Multiple Document Interface. SQL = Structured Query Language. DBA = Database Administrator. OLE = Object Linking and Embedding.

اردو میں سمجھیں

آپشن II: فارم کے نام cmd، txt، lbl رکھیں۔ Val سے جمع کریں۔ Access کے سوال C والے کاغذ جیسے ہی ہیں۔

MCQs

1. ADO stands for:
 - a. ActiveX Data Objects
 - b. Access Data Only
 - c. All Data Output
 - d. Array Data Object

Key: 1=a

Long questions (board Section C)

1. Design on paper a VB form with two text boxes, four command buttons (+ − × ÷) and a label for the result. Write the four Click events.